2 P-N Junction Diodes

INTRODUCTION

- A P-N junction diode is a two-terminal device that is formed by P and N-type semiconductors. P-type materials are doped in one terminal, while N-type materials are doped in another. The P-N junction is the primary unit of semiconductor diodes. The core function of a semiconductor diode is to enable the electrons to flow entirely in one direction only.
- Figure 2.1 shows the diode sign, with the P-type and N-type areas referred to as anode and cathode, respectively. The arrowhead indicates the traditional current flow direction, which is the direction in which hole flow occurs.



Fig. 2.1 Diode

- When a side of the junction is doped with acceptor impurities (trivalent atoms) in the P-region, but the other side is doped with donor impurities (pentavalent atoms) in the N-region, the P-N junction is formed.
- The junction, also known as the semiconductor barrier, is the area where these N-type and P-type materials are fused.
- In the P-region, the majority of charge carriers are holes, whereas, in the N region, the majority of charge carriers are electrons.
- When the junction is constructed, the holes in the P-region diffuse into the N-region, leaving the negative charge in the P-region, which recombines with electrons to form negatively charged ions in the P-region.
- The application of this unidirectionality in a diode is rectifiers, which are basically AC to DC converters.
- The basic requisite for further topics:
  1) Poisson's equation.
  2) Electric field and voltage definitions.

Electric field and voltage relationship:

- The voltage is most often given to the conductor as energy. Whenever the voltage is given to the conductor, it will experience force around it in the free space.
- For example: Take capacitor plates of an area of cross-section 'A' and separated by 'd' distance.

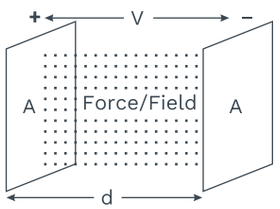


Fig. 2.2 Capacitor Plates

The voltage between the plates of the capacitor produces the invisible force or field.

- The electric field is the stress/force developed in length 'd' or the voltage spread over the 'd' length is regarded as the electric field.

Mathematically, $\frac{V}{d} = E$

- If the length is longer, then the stress will be less, and for the shorter length, the stress will be more.
- The electric field is defined as the rate of change of voltage over the length.

The SI unit of the electric field is Volts/meters.

- Electric field intensity or strength of the field is the rate of decrease of voltage per unit length or voltage stress over a length.

POISSON'S EQUATION

- As we know that the Poisson equation is:

$$\nabla \cdot \vec{D} = \rho_v$$

where ρ_v = Bulk charge/Volume charge density (C/m^3) and $\rho_v = ne$

n = Number of electrons per unit volume

e = Charge of the electron/carrier

If N-type semiconductor,

$\rho_v = N_D e$

For P-type semiconductor,

$\rho_V = N_A e$

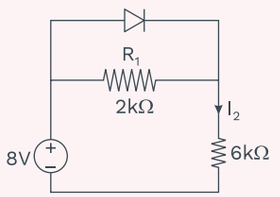
- $\vec{D}$ is the strength of the field or called as density and $\vec{E}$ is the electric field intensity

where $\vec{D} = \in \vec{E}$

∇ is the rate of change in space

Rack Your Brain

The diode has V_{0N} = 0.7V but is ideal in nature otherwise.



The magnitude of I_2 (mA) will be_____.

Thus, $\nabla \cdot \vec{D} = \rho_v$ is expressed as the rate of change of the strength of the field depending on the charge concentration.

- This is also regarded as the Gauss law and Maxwell's first equation.

We know that:

$$\nabla \cdot \vec{D} = \rho_v$$

$$\vec{D} = \in \vec{E}$$

$$\vec{E} = -\nabla V$$

By substitution, we get:

$$\nabla \cdot \left(\in \vec{E}\right) = \rho_v$$

$$\nabla . \left(\in \left(-\nabla V\right)\right) = \rho_v$$

$$\nabla^2 V = -\frac{\rho_v}{\in}$$

The above equation is regarded as Poisson's equation.

- The minus sign here denotes the decreasing voltage in the electric field direction.
- The electric field is always positive to a negative direction or higher potential to lower potential direction, thus the decreasing voltage direction.
- The Poisson's equation is the relation between the voltage developed because of any charge.

JUNCTION FORMATION AND DEPLETION REGION

- When the P-N junction forms, the holes on the P-side and electrons on the N-side will undergo a diffusion process, while both regions being electrically neutral.
- Because of the concentration difference initially, the diffusion will take place gradually.

Previous Years' Question

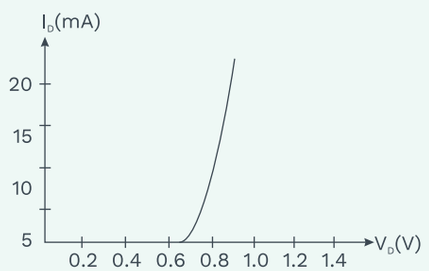
The piecewise linear equivalent circuit model for the diode shown in the following figure comprises of:

a) a battery voltage of 0.8V, the resistance of 10Ω and an ideal diode.

b) a battery voltage of 0.8V, the resistance of 11Ω and an ideal diode.

c) a battery voltage of 0.6V, resistance 10Ω and an ideal diode.

d) cannot be determined from given data.



The Voltage difference between points A and B = 50 mV.

This diffusion will result in junction formation, and as soon as the junction formation is done, the holes and electrons will be accumulated around the junction.

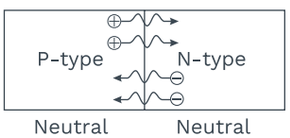


Fig. 2.3 Diffusion in P-N Junction

Thus, the diffusion of majority charge carriers results in junction formation.

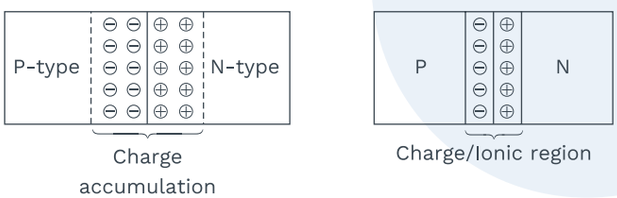


Fig. 2.4 Junction Formation

The region of charges/ions has no carriers, and these charges or ions are immobile.

- The effect of diffusion is the formation of ions or the extra region that stops the exchange of the carriers.
- Diffusion of majority carriers at the junction will cause the immobile ions and a region containing zero carriers to have a strong electric field.

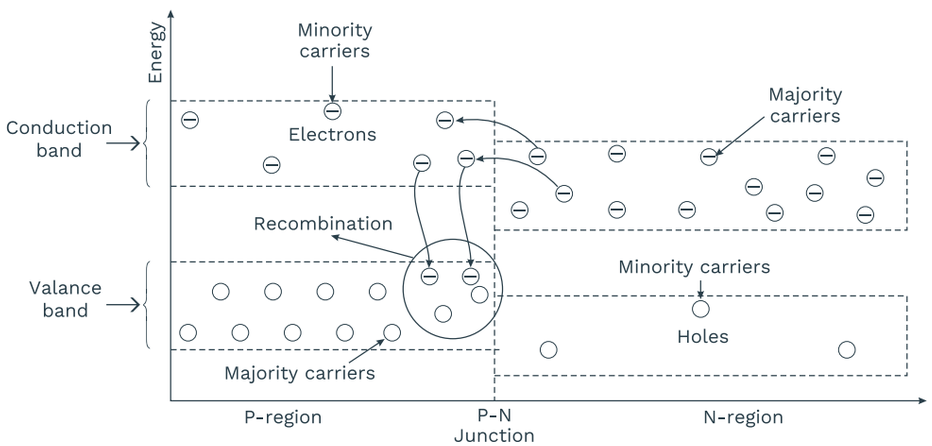


Fig. 2.5 Junction Formation

Rack Your Brain

Consider the wire at home is supplied by 230V. What will happen then?

a) The Wire will be damaged.
b) The fuse at the source will trip.
c) None of the above.
d) Both A) and B).

- The electrons from the N-region conduction band will move to the P-region conduction band because of concentration differences but ultimately goes to the valence band of the P-region and gets involved in the recombination process.
- Due to this, the permanent negative charge is on P-side, and the positive charge is on the N-side.

Note:

Generation of electron-hole pair will always require energy, whereas the energy is released during the recombination region.

Therefore, there are three-regions in a diode:
P-region, N-region and the depletion region.

- The diffusion and the subsequent deposition of carriers result in the region of zero carriers or depleted of carriers.
- The region with depleted carriers is regarded as the depletion region or space charge region.

As we know, when an electron moves from E_v (Valence Band) to E_c (Conduction Band), then two carriers will be generated.

But when an electron moves from E_c to some other levels, ions will be formed.

Abrupt PN junction:

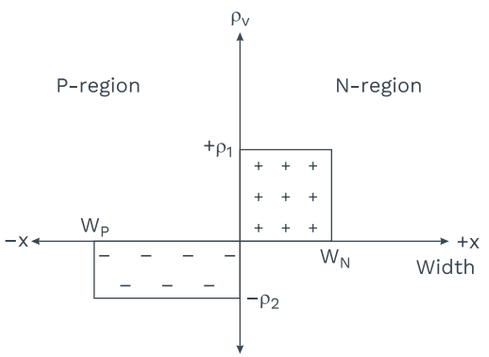
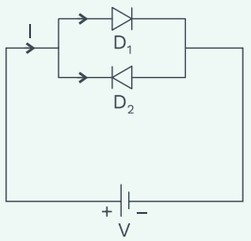


Fig. 2.6 Space Charge Density in Uniformly Doped Abrupt PN Junction

Previous Years' Question

In the circuit shown below, the current-voltage relationship when D_1 and D_2 are identical, as given by: (Assume Ge diodes)



a) $V = \frac{kT}{q}\sinh\left(\frac{I}{2}\right)$

b) $V = \frac{kT}{q}\ln\left(\frac{I}{I_0}\right)$

c) $V = \frac{kT}{q}\sinh^{-1}\left(\frac{I}{2}\right)$

d) $V = \frac{kT}{q}\left[\exp(-I) - 1\right]$

- The charge density $\rho_1 = N_D e$ and $\rho_2 = N_A e$

The width of the depletion region on the P-side is defined as W_P. Similarly, W_N is the width of the depletion region on the N-side.
Thus,
The total width of the depletion region

$W = W_P + W_N$

- The relation between the charge densities ρ_1 and ρ_2 will be:

$\rho_1 A W_N = \rho_2 A W_P$

where A = area
ρ_1 = charge density on N-side
ρ_2 = charge density on P-side

$\therefore \quad N_D e\, A W_N = N_A e A W_P$

Thus, $N_D W_N = N_A W_P$ (Charge Neutrality)

This junction where charge ρ_v or ρ_1 or ρ_2 is constant with the distance is called step-graded junctions.
This step-graded junction is developed due to alloy junctions or uniform doping in P and N regions.

Linearly graded junction:

- The manufacturing construction and dopants also sometimes cause a different form of a junction.
- The possibility of other types of the junction will be understood by the following graph.

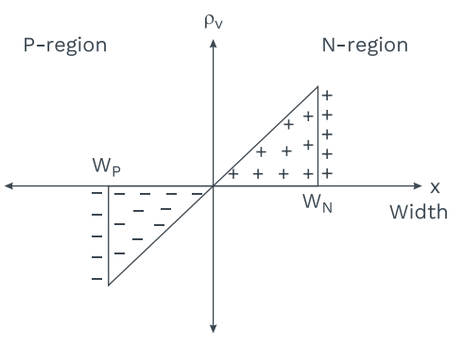


Fig. 2.7 Space Charge Density in Linearly Graded PN Function

Previous Years' Question

A combination of two diodes connected in parallel when compared to a single diode can withstand:

a) twice the value of the peak inverse voltage
b) twice the value of the maximum forward current.
c) a larger leakage current.
d) twice the value of cut-in voltage.

From figure,

$\rho_v \propto x$

$\rho_v = kx$

or $\begin{cases} \rho_v = +kx, & \text{for } N-\text{side} \\ \rho_v = -kx, & \text{for } P-\text{side} \end{cases}$

This is referred to as linearly graded or diode or grown junction.

- Also, the N-region and P-region do not have uniform charges. They have distance-increasing charge concentrations.
- The ions accumulation is not maximum at the junction; it is gradually increasing away from the junction.

Step-graded diode:

In a step-graded diode, we know that:

$\rho_1 = N_D e$

$\rho_2 = N_A e$

and $\nabla^2 V = -\frac{\rho_V}{\in}$

Applying the above Poisson equation for N-side and P-side.

- The electric field and voltages for a step-graded diode for N-region:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} = -\frac{N_D e}{\in} = -\frac{\rho_1}{\in}$$

The above equation is for one dimension or the length is along with the diode.

Integrating on both sides:

$$\frac{\partial V}{\partial x} = E = -\frac{N_D e \cdot x}{\in} + K$$

Here, the electric field is linearly increasing with distance.

For P-region:

$$\frac{\partial V}{\partial x} = E = -\frac{N_A e \cdot x}{\in} + K$$

Rack Your Brain

If the combination of two diodes is connected in series & when it is compared to a single diode that can hold _____ the value of peak the inverse voltage or cut-in voltage.

In general, the electric field is directed in the left direction.

- At x = W_N, the E = 0 and x = 0, E is maximum, but again moving in a leftwards direction, decreases back to zero.

Since the electric field moves from positive charge to negative charge.

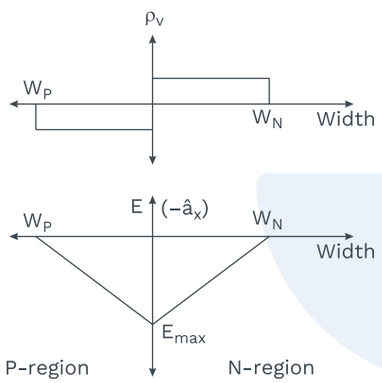


Fig. 2.8 For Linearly Graded Diode

- This electric field is drawn in the third and fourth quadrant because the direction is to the left side.

On reversing the N-region and P-region, the graph will shift to the first and second quadrants.

Since the charge is constant:

ρ_v (x) = constant

= $N_D e$ or $N_A e$

and E(x) is linear:

$E(W_P) = E(W_N) = 0$

$E(0) = E_{max}$

- The maximum electric field (E_{max}) will be:

$$E_{max} = \frac{N_D e W_N}{\in} = \frac{N_A e W_P}{\in}$$

The direction is from the N-region to P-region.

Rack Your Brain

Can you realize an inductance in an IC?

At x = 0, V = 0 (At interface of P and N side)

- The voltage peak value on the P-region is given by:

$$V_{0P} = -\frac{N_A e W_P^2}{2 \in} \qquad \{\text{Maximum at } W_P\}$$

On the N-region

$$V_{0N} = -\frac{N_D e W_N^2}{2 \in} \qquad \{\text{Maximum at } W_N\}$$

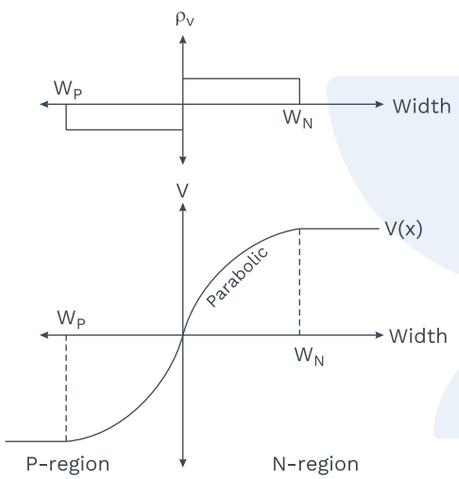
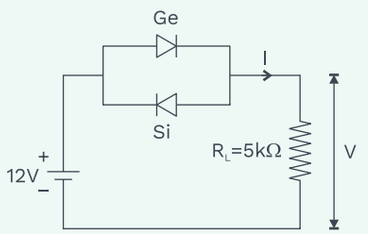


Fig. 2.9 Electric Potential Through Space Charge Region

Previous Years' Question

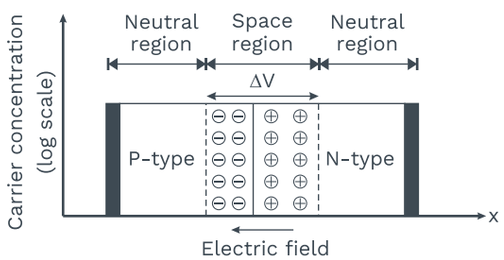
Calculate the voltage (V) across R_L. In the circuit below:



The junction opposing voltage is:

$$V_0 = V_{0N} - V_{0P} \text{ (Cut-in voltage)}$$

$$\therefore \quad V_0 = \frac{N_D e W_N^2}{2 \in} + \frac{N_A e W_P^2}{2 \in}$$



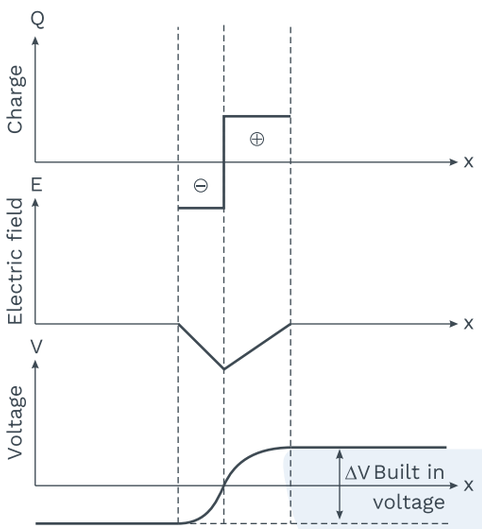


Fig. 2.10 For Step Graded Diode

- The direction of an electric field is in the direction of decreasing voltage.

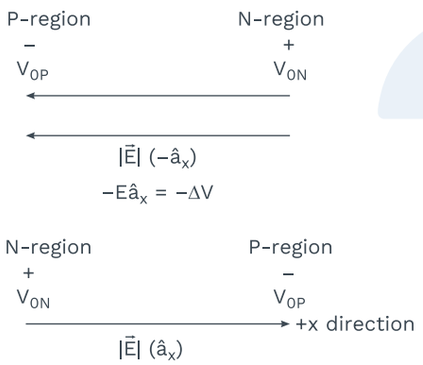


Fig. 2.11 Direction of Electric Field

- Thus, at x = 0, $\vec{E} = \vec{E}_{max}$

$$\vec{E} = 0 \begin{cases} \text{At} & x = W_N \\ & x = W_P \end{cases}$$

At any distance (x),

$$\vec{E}(x) = \frac{eN_D}{\in}(x - W_N)\hat{a}_x \qquad (\text{for N} - \text{region})$$

Rack Your Brain

If V_F = 5V and V_R = –3V, then will storage time increase or decrease?

Correspondingly:

$$\vec{E}(x) = -\frac{eN_A}{\in}(x + W_P)\hat{a}_x \qquad \text{(for P-region)}$$

Cut in voltage and width of the depletion region:
We know that:

$$V(x) = -\frac{eN_D x^2}{2 \in} = -\frac{eN_A x^2}{2 \in}$$

The potential difference:
At x = 0 , V(x) = 0
At x = W_N , V(x) = V_{0N}
At x = W_P , V(x) = V_{0P}
where V(x) = cut in voltage or barrier voltage.

- This barrier potential/opposite voltage is developed because of electric field present is referred to as cut-in voltage or barrier voltage which is possessed by the diode without any application of external voltage.

Example 1:
Consider a capacitor plate having –Q and +Q on each plate. Is the voltage between plates uniform, and is the electric field constant?

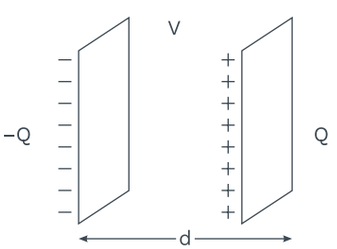


Fig. 2.12 Parallel Plate Capacitor

Sol:
Inside the capacitor, there is the region or array of plates, and $\vec{E}$ varies linearly with x.

$E \propto x$

and $V \propto x^2$

For single-plate capacitors:

$\vec{E} = kx$ (Constant)

Previous Years' Question

The switching speed of the p^+n junction (having a heavily doped p region depends primarily on):

a) The mobility of minority carriers in the p^+-region.
b) The lifetime of minority carriers in the p^+-region.
c) The mobility of majority carriers in the n-region.
d) The lifetime of majority carriers in the n-region.

and $V = kx^2$

Hence, it is concluded that the electric field is constant, and the voltage is uniform.

- Open circuit voltage or built-in voltage or cut-in voltage or barrier potential because of the depletion region is expressed as:

$$V_0 = \frac{eN_D W_N^2}{2 \in} + \frac{eN_A W_P^2}{2 \in}$$

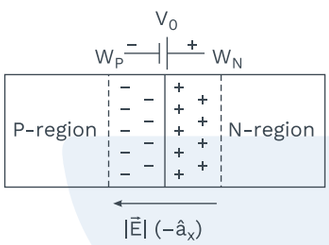


Fig. 2.13 Barrier Potential

- Total width (W) of depletion region will be:

  $W = W_N + W_P$

And the relation between cut-in voltage and width of depletion region can be understood by:

$$N_D W_N = N_A W_P$$

The cut-in voltage is:

$$V_0 = \frac{eN_D W_N^2}{2 \in} + \frac{eN_A W_P^2}{2 \in}$$

$$V_0 = \frac{e}{2 \in}\left[N_D W_N W_N + N_A W_P W_P\right]$$

$$V_0 = \frac{e}{2 \in} N_D W_N \left(W_N + W_P\right) \qquad \{\because N_D W_N = N_A W_P\}$$

$$V_0 = \frac{eN_D}{2 \in} \cdot W_N \cdot W \qquad \{\because W = W_P + W_N\}$$

and we know that:

$$N_D W_N = N_A W_P$$

$$N_D W_N = N_A (W - W_N)$$

$$N_D W_N + N_A W_N = N_A W$$

Previous Years' Question

In a junction diode:

a) The depletion capacitance increases with an increase in the reverse bias.

b) The depletion capacitance decreases with an increase in the reverse bias.

c) The depletion capacitance increases with an increase in the forward bias.

d) The depletion capacitance is much higher than the depletion capacitance when it is forward biased.

Thus, $W_N = \frac{N_A W}{N_D + N_A}$

Now, after substituting, we get:

$$V_0 = \frac{eN_D}{2 \in}\left(\frac{N_A \cdot W}{N_D + N_A}\right)(W)$$

or $V_0 = \frac{eN_D}{2 \in_0}\frac{N_A \cdot W^2}{(N_D + N_A)}$

The above expression gives a relation between the width depletion region and the doping concentration for cut-in voltage.

- It is concluded that the width of the depletion region versus doping concentration are related as:

Since, we know that:

$$V_{0P} = \frac{eN_A W_P^2}{2 \in} \text{ (for P-region)}$$

$$V_{0N} = \frac{eN_D W_N^2}{2 \in} \text{ (for N-region)}$$

Thus, $\text{Width} \propto \frac{1}{\sqrt{\text{Doping}}}$

and also we know that:

$$N_D W_N = N_A W_P$$

or $\frac{N_D}{N_A} = \frac{W_P}{W_N}$ {Inverse relation}

- If a side has higher doping, then it will have a smaller depletion region.

Note:

The above derivation is mostly used for alloy junction or step-graded diode, or uniform doping.

where $\rho_{VN} = N_D e$ and $\rho_{VP} = N_A e$

- If the diode is a grown junction or linearly graded diode.

where

$\rho_V \neq N_D e$

$\rho_V \neq$ Constant

$\rho_V = kx$ (or $\rho_V \propto x$)

then, $E \propto x^2$ and $V \propto x^3$

LAW OF JUNCTION

- When the diffusion of the carriers occurs across the junctions, it results in the counter electric field.

The current density (J) will be:

$J_{Diffusion} = J_{Drift}$ of electric field

A balance/equilibrium state is achieved, or a steady state is achieved.

$$eD_P \frac{dP}{dx} = Pe\mu_P \frac{dV}{dx}$$

We can say that the diffusion stops because of drifting back of the electric field, but this is not the case.

The electric field is developed by the ions.

$\therefore$ The above equation can be re-written as:

$$\frac{D_P}{\mu_P} . \frac{dP}{P} = dV$$

Integrating on both sides, we get:

$$V_T \left[\ln P\right]_N^P = V_{across\ junction}$$

Since, $\frac{D_P}{\mu_P} = V_T$

$$\therefore \quad V_T \left[\ln P\right]_{P_N}^{P_P} = V_0 \quad \text{(Across junction)}$$

where

P_P = P-region holes

P_N = N-region holes

V_0 = Barrier potential

$$V_0 = V_T \ln \frac{N_A}{\left(\frac{n_i^2}{N_D}\right)}$$

$$\therefore \quad V_0 = V_T \ln \left(\frac{N_A N_D}{n_i^2}\right)$$

The above expression provides a relation between the doping and cut-in voltage.

Rack Your Brain

How is the width related to applied voltage in the linearly graded diode?

The cut-in voltage will be:

$$V_0 = V_T \ln\left(\frac{\text{Majority carriers in region 1}}{\text{Minority carriers in region 2}}\right)$$

or $V_0 = V_T \ln\left(\frac{N_N}{N_P}\right)$

- If region 1 is the N region and region-2 is P-region.
- The cut-in voltage is always doping concentration-independent.

Example 2:

Consider a silicon diode with $n_i^2 = 1.5 \times 10^{10} cm^{-3}$, $N_A = 10^{16} cm^{-3}$ and $N_D = 10^{16} cm^{-3}$. Calculate the cut-in voltage.

a) 0V **b)** 0.3V **c)** 0.8V **d)** 1V

Sol:

Given that a silicon diode has:

$n_i^2 = 1.5 \times 10^{10} cm^{-3}$

$N_A = 10^{16} cm^{-3}$

$N_D = 10^{16} cm^{-3}$

The cut-in voltage will be:

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$\therefore \quad V_0 = 26 \times 10^{-3} \times \ln\left(\frac{10^{16} \times 10^{16}}{1.5 \times 10^{10}}\right)$$

Hence, for practical doping of N_A or N_D with lower n_i or higher as 10^{19}.

Then also, the V_0 is not that much affected due to doping.

$\therefore$ Practically, the cut-in voltage for silicon is $V_0 = 0.7V$ and for germanium is $V_0 = 0.3$ V.

Previous Years' Question

In a uniformly doped abrupt P-N junction, the doping level of the n-side is four times, the doping level of the p-side, the ratio of the depletion layer width of n-side versus p-side is:

a) 0.25 **b)** 0.5
c) 1.0 **d)** 2.0

Example 3:

Consider an abrupt PN junction at room temperature. The depletion region width W_N on the N-region of the junction is 0.2 μm, and the permittivity of silicon is 1.044×10^{-12} F/cm.

The peak electric field (in kV/cm) at the junction approximately is:

a) 306.5 **b)** 30.65 **c)** 3.65 **d)** 36.5

Sol:

Given that:

The width of the depletion region on the N-side is

$W_N = 0.2\ \mu m$

and since the doping on the P-region is heavy.

Thus, the width of the depletion region at that side will be smaller and considered a negligible width.

$\therefore \quad W_P \simeq 0$

$\therefore$ The total width of the depletion region is:

$W = W_N = 0.2\ \mu m = 0.2 \times 10^{-4}$ cm (Given)

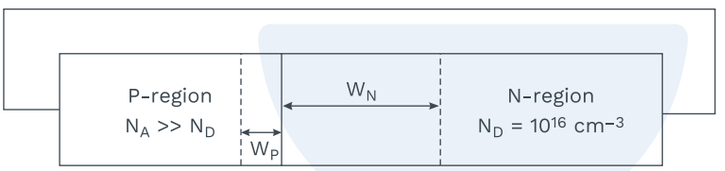


Fig. 2.14 Depletion Region Width of Abrupt PN Junction

The peak electric field (E_{max}) will be:

$$E_{max} = \frac{eN_D}{\in} W_N$$

$$E_{max} = \frac{1.6 \times 10^{-19} \times 10^{16} \times 0.2 \times 10^{-4}}{1.044 \times 10^{-12}}$$

$E_{max} = 30.65$ kV/cm

Hence, the correct option is b).

Example 4:

The doping concentrations of an abrupt P-N junction are $N_A = 9 \times 10^{16}$ cm^{-3} and $N_D = 1 \times 10^{16}$ cm^{-3}. If P-N junction is reversed biased, and the total width of the depletion is 3 μm. Then the width on the P side is:

a) 0.3 μm
b) 0.3 μcm
c) 30 m
d) 0.3 mm

Previous Years' Question

The small-signal capacitance of an abrupt p^+n junction is 1nF/cm^2 at zero bias. If the built-in voltage is 1 volt, the capacitance at the reverse bias voltage of 99 volts in (nF/cm^2) is:

a) 10 **b)** 0.1
c) 0.01 **d)** 100

Sol:

Given that:

Doping concentration on P-region and N-region are:

$N_A = 9 \times 10^{16}\ cm^{-3}$

$N_D = 1 \times 10^{16}\ cm^{-3}$

Total depletion region width (W) = 3 μm

$= 3 \times 10^{-4}$ cm

Since, we know:

$N_D W_N = N_A W_P$

$N_D (W - W_P) = N_A W_P$

$N_D W - N_D W_P = N_A W_P$

$N_D W = (N_A + N_D) W_P$

$$W_P = \frac{N_D W}{(N_A + N_D)}$$

$$W_P = \frac{1 \times 10^{16} \times 3 \times 10^{-4}}{(9 \times 10^{16} + 1 \times 10^{16})}$$

$$W_P = \frac{10^{16} \times 3 \times 10^{-4}}{10 \times 10^{16}}$$

$W_P = 3 \times 10^{-5}$

$W_P = 0.3\ \mu m$

Hence, the correct option is a).

Rack Your Brain

What feature makes the Zener diode a special diode?

Example 5:

Which of the following statement is incorrect?

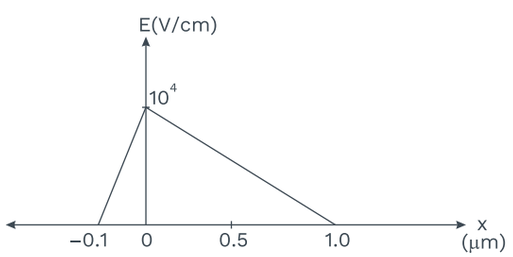


Fig. 2.15 Electric Field vs Distance

a) The right side of the junction is P-type and the left side is N-type.

b) The potential difference across the depletion region is 700 mV.

c) Both the N-type and P-type depletion regions are uniformly doped.

d) The P-type region has a doping concentration of $10^{15}/cm^3$, and then the doping concentration in the N-type region is $10^{16}/cm^3$.

Sol:

The distance versus the electric field, as shown in the given figure depicts that the:

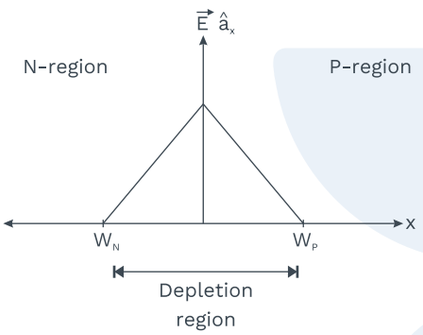


Fig. 2.16 Electric Field vs Distance

∴ Option a) is correct and states that the left side is N-type and the right side is P-type. Hence, option a) is correct.

- Also, both the P-type and N-type depletion are uniformly doped because the graph has straight lines on the left and right sides.

But they are not equally doped but uniformly doped. Option c) is also correct.

The potential difference across the depletion of the region is expressed as:

$$V = \int \vec{E} \cdot d\vec{l} = E \times l$$

$$\therefore \quad V = \left(\frac{1}{2} \times 0.1 \times 10^4 + \frac{1}{2} \times 1 \times 10^4 \right) \times 10^{-4}$$

$$V = 0.05 + 0.5 = 0.55 \text{ V}$$

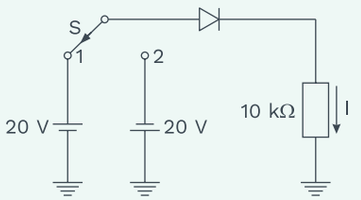
$$V = 550 \text{ mV}$$

Hence, option b) is incorrect.

Previous Years' Question

Referring to the below figure, the switch S is in position 1 initially, and steady-state conditions exist from time t = 0 to $t = t_0$. At $t = t_0$, the switch is suddenly thrown into position 2.

The current I (in mA) through the 10 kΩ resistor at $t = t_0$ is _____.



- As we know:

$$E_{max} = \frac{eN_D W_N}{\in} = \frac{eN_A W_P}{\in}$$

$$10^4 = \frac{1.6 \times 10^{-19} \times N_D \times 0.1 \times 10^{-4}}{8.85 \times 10^{-12}}$$

$$N_D = \frac{8.85 \times 10^{-12} \times 10^4}{1.6 \times 10^{-19} \times 0.1 \times 10^{-4}}$$

$$N_D = 55.31 \times 10^{15}\ cm^{-3}$$

This shows that the doping concentration is approximately 10^{15} and 10^{16} cm^{-3}. Option d) is also correct.
Thus, only option b) is incorrect.

Example 6:
A region of silicon is devoid of electrons and holes, having an ionized donor density of 10^{17} cm^{-3}.
The electric fields at x = 0 and x = L are 0 V/cm and 50 kV/cm in the positive x-directions.
The electric field is zero at all points in the y and z directions.
Given that:
$\in_r$ = 11.7 for silicon $\in$ = 8.85 × 10^{-14} F/cm
The value of L (in ncm) will be:

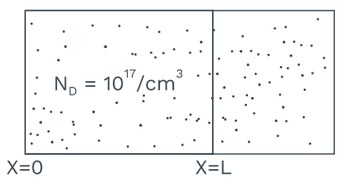


Fig. 2.17 A Region in Silicon Wafer

a) 3.235
b) 3.300
c) 3235.78
d) 32.3578

Sol:
Given that:

$N_D = 10^{17}\ cm^{-3}$

E = 0 V/cm at x = 0

E = 50 kV/cm at x = L

Previous Years' Question

The built-in potential (diffusion potential) in a P-N junction:
a) Is equal to the difference in the Fermi levels of the two sides and expressed in volts.
b) Increases with the increase in the doping levels of the two sides.
c) Increases with the increase in temperature.
d) Is equal to the average of the Fermi levels of the two sides.

ϵ_r = 11.7 for silicon

$\epsilon_0 = 8.85 \times 10^{-14}$ F/cm

The electric field at L will be expressed as:

$$E(L) = \frac{eN_D L}{\epsilon}$$

Since, x = 0 varies upto x = L
and E = 0 to 50 kV/cm
Substituting values in above equation:

$$50 \times 10^3 = \frac{1.6 \times 10^{-19} \times 10^{17} \times L}{11.7 \times 8.85 \times 10^{-14}}$$

$L = 3235.78 \times 10^{-14+5}$

$L = 3235.78 \times 10^{-9}$ cm

L = 3235.78 ncm

The value of L is 3235.78 ncm.
Hence, the correct option is c).

ENERGY BAND DIAGRAM

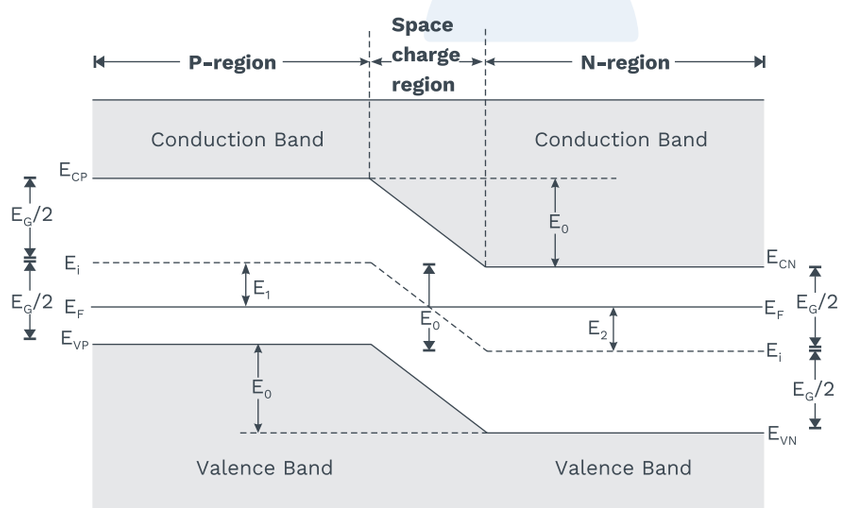


Fig. 2.18 Energy Band Diagram

As we know, that cut-in voltage is given by:

$$V_0 = V_T \ln\left(\frac{N_D N_A}{n_i^2}\right)$$

Also, $V_0 = \frac{e}{2 \in}\left(\frac{N_D N_A}{N_D + N_A}\right) W^2$

- To understand the energy band diagram, consider a single crystal or bulk material.

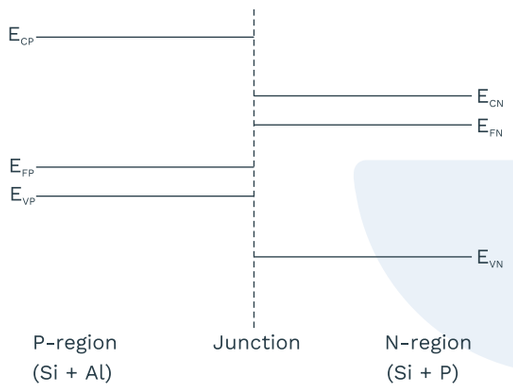


Fig. 2.19 Energy Band Diagram

- Before the junction forms, the above shows the ionization phase.

But after the formation of the junction, the holes from the P-region (left side) will move to the right side.

- Similarly, it is observed that the electrons will move to the left side from the right side.

The conduction band in the P-region is certainly higher than the conduction band in the N-region, and correspondingly, the Fermi level in the N-region is higher than that of the P-region. This is why the electrons will move from right to left.

The positive charges in P-region and negative charges in N-region also depict a higher value/ level.

Thus, the electrons will move, as shown in the figure,

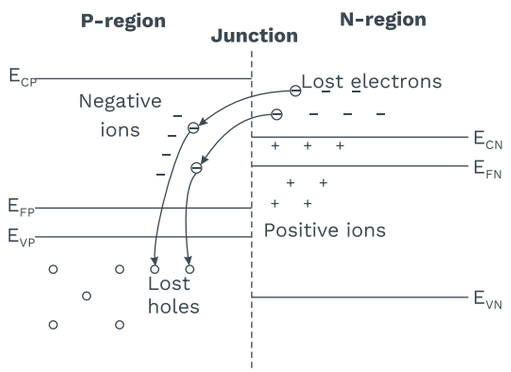


Fig. 2.20 Movement of Electrons

The positive ions region is developed around the right side of the junctions.

- Similarly, the region of the negative ion is produced near the left side of the junction because of the recombination process. The above process is repeated until the Fermi-level shifts upwards (as electrons are coming from the right side), and it is losing its P-nature.

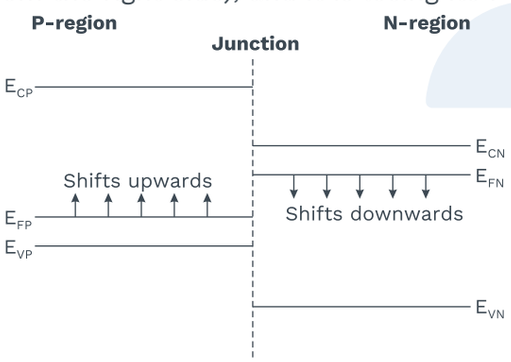


Fig. 2.21 Shifting of Fermi Levels

- The n-nature on the N-region is lost since holes are shifting towards the right side.

This shifting upwards and downwards will stop when the Fermi level on both sides is on the same level, and also, the formation of the depletion region stops.

Here, P-region and N-region are considered silicon only but the atoms added are different for both sides; therefore, the energy levels are different.

- With P-region and N-region, the energy band is shown in figure (2.18); it is concluded that $E_{FP} < E_{FN}$ and the electrons move from higher to lower energies or can say that N-region to P-region.

Previous Years' Question

The diffusion potential across a P-N junction:

a) Decreases with increasing doping concentration.

b) Increase with decreasing band gap.

c) Does not depend on doping concentration.

d) Increases with an increase in doping concentration.

Rack Your Brain

Why is heavy doping required in the Zener diodes?

This flow of electrons is regarded as diffusion, and this diffusion results in the region of the ion formation electric field and the potential difference across the junction.

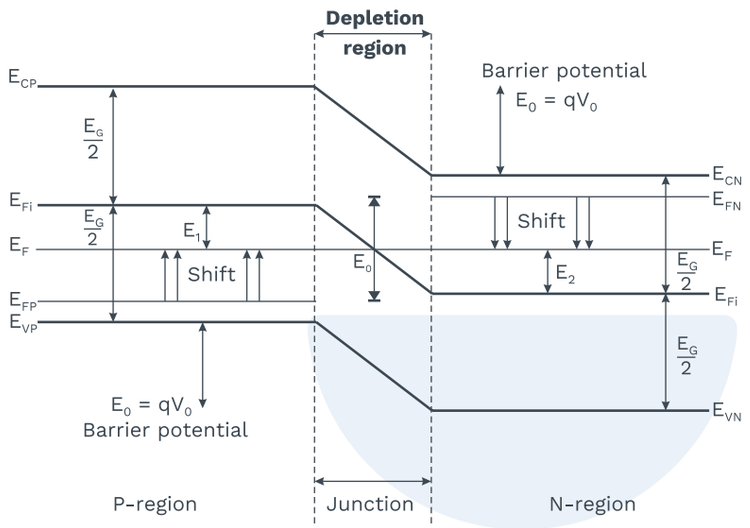


Fig. 2.22 Equivalent Energy Band Diagram

E_F = Fermi-level after junction formation.
E_{FP} = Fermi-level before junction formation on P-side.
E_{FN} = Fermi-level before junction formation on N-side.
E_{Fi} = Intrinsic fermi-level.

Note:

- If the construction is made up of N-type and intrinsic, then the energy band diagram will be as follows:

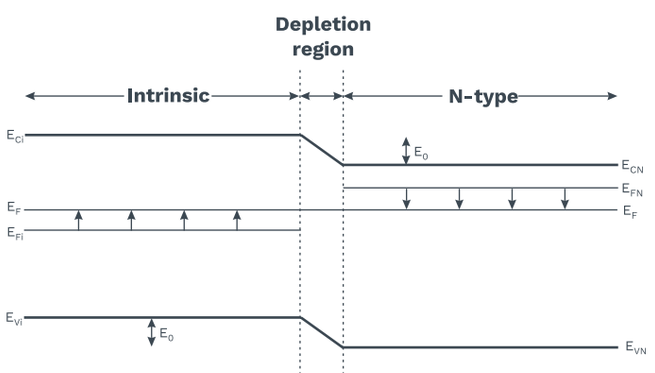


Fig. 2.23 Energy Band Diagram of Intrinsic and N-Type Junction

Here, the barrier potential (V_0) will be less and can result in an advantage, but the reverse current will flow and can disturb the other properties.

- In microwave devices, we use this kind of arrangement (Example: PIN diode). Similarly, for intrinsic and P-type, the situation will be the same as above and is described in the following diagram.

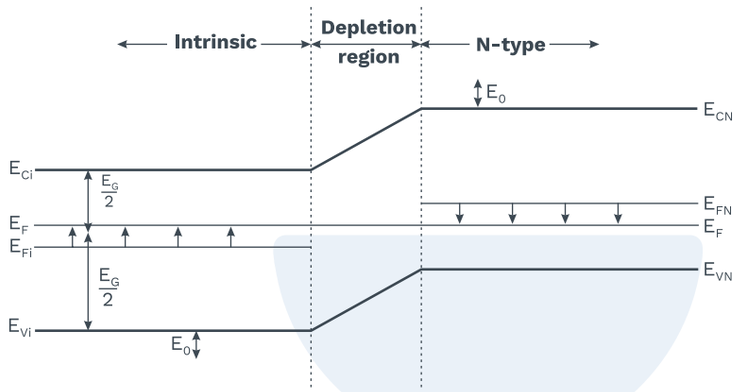


Fig. 2.24 Energy Band Diagram of Intrinsic and P-Type Junction

- Now, we know that:

$$E_0 = E_1 + E_2$$

where E_1 = Fermi-level on the P-region shifted from intrinsic.
E_2 = Fermi-level on the N-region shifted from the intrinsic.
E_G = Energy band gap.
Thus, cut-in voltage ($E_0 = qV_0$) can be derived from E_1 and E_2.
Therefore,

$$E_1 = \frac{E_G}{2} = -KT \ln\left(\frac{N_V}{N_A}\right)$$

Here, on P-side

The shift is the $KT \ln\left(\frac{N_V}{N_A}\right)$ from the E_{VP}

$\therefore$ Similarly, on N-side –

$$E_2 = \frac{E_G}{2} - KT \ln\left(\frac{N_C}{N_D}\right)$$

Here, $KT \ln\left(\frac{N_C}{N_D}\right)$ = Shift from E_{CN}

Previous Years' Question

The static characteristic of an adequately forward-biased P-N junction is a straight line. If the plot is of:

a) (log I) vs (log V)
b) (log I) vs (V)
c) (I) vs (log V)
d) (I) vs (V)

$$\therefore \quad E_0 = E_1 + E_2$$

$$E_0 = E_G - KT \ln\left(\frac{N_C N_V}{N_A N_D}\right)$$

Since we know that:

$$n_i = n = N_C e^{(E_C - E_F)/KT}$$

$$n_i = p = N_V e^{(E_F - E_V)/KT}$$

Also,

$$n_p = n_i^2 = N_C N_V e^{-\frac{E_G}{KT}}$$

$$\therefore \quad E_G = -KT \ln\left(\frac{n_i^2}{N_C N_V}\right)$$

$$E_G = KT \ln\left(\frac{N_C N_V}{n_i^2}\right)$$

Now substitute the value of E_G in the cut-in voltage (E_0) equation; we get:

$$E_0 = E_G - KT \ln\left(\frac{N_C N_V}{N_D N_A}\right)$$

$$E_0 = KT \ln\left(\frac{N_C N_V}{n_i^2}\right) - KT \ln\left(\frac{N_C N_V}{N_D N_A}\right)$$

$$E_0 = KT \ln\left(\frac{N_D N_A}{n_i^2}\right) \quad \text{(Joules)}$$

$$\text{Or} \quad V_0 = \frac{KT}{e} \ln\left(\frac{N_D N_A}{n_i^2}\right) \quad \text{(Volts)}$$

- For a conductor, we know that:

The conduction band and valance band are nearer to each other, or the energy band gap is less.

It allows the easy flow of electrons from the valence band to the conduction band.

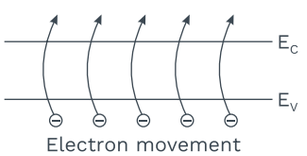


Fig. 2.25 Electron Movement

- In the case of dielectric, there is a huge gap between the conduction band and the valence band. Thus, there will be a negligible flow of electrons.

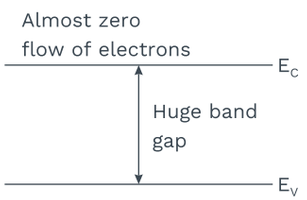


Fig. 2.26 For Dielectric

- In the case of intrinsic semiconductors, there are few movements of electrons, and the energy band gap is moderately distant. It has two level movements.



Fig. 2.27 For Intrinsic Semiconductor

Thus, it is concluded that there will be movements of charges in the conductor and intrinsic semiconductor but not in the dielectric material. The diode shows different properties from all the above mentioned three cases.

- In the diode, we will observe that:

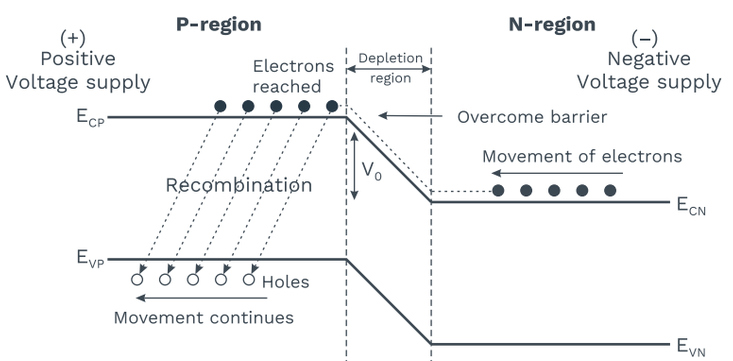


Fig. 2.28 Movement of Electrons in Diode

Previous Years' Question

A P-N junction in series with a 100 ohms resistor is forward biased so that a current of 100 mA flows. If the voltage across this combination is instantaneously reversed at t = 0, the current through the diode is approximately given by:

a) 0 mA **b)** –100 mA
c) 200 mA **d)** 50 mA

- If we supply negative voltage to N-region and positive voltage to the P-region. The electrons will be pushed to the left side from the right side, and the current will be produced.

And to have movement from E_{CN} to E_{CP}, the minimum voltage required is equal to the built-in potential.

Because of this application of V_0 (minimum voltage), it will overcome the barrier.

As soon as the electrons move from E_{CN} to E_{CP}, they will be sustained there and will ultimately fall to the valence band and get recombined with holes present there.

- Without external supply, the diode has less flow of electrons.

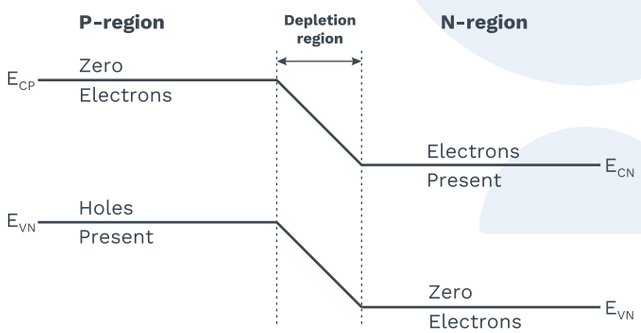


Fig. 2.29 Without External Supply

- If P-region is given positive voltage and N-region is given negative voltage, then the arrangement is referred to as forwarding bias.

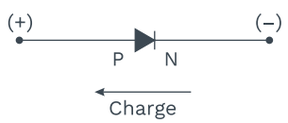


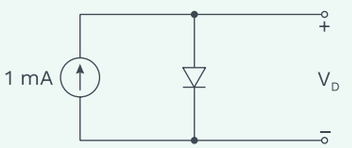
Fig. 2.30 P-N Diode in Forward Biasing

The positive charge moves from left to right side, and the negative charge moves from right to left side.

Previous Years' Question

In the given figure, a silicon diode carries a constant current of 1 mA.

When the temperature of the diode is 20°C. V_D is found to be 700 mV. If the temperature rises to 40°C. V_D becomes approximately equal to:



a) 740 mV **b)** 660 mV
c) 680 Mv **d)** 700 mV

- The graph between diode current and diode voltage is explained as follows:

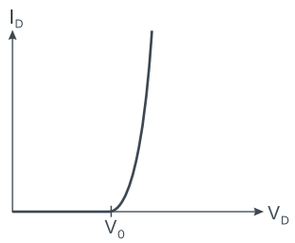


Fig. 2.31 V-I Characteristics of Diode in Forward Bias

It is observed that there will be no flow of current until V_0 .
After V_0, the rate of current flow grows exponentially.

- Suppose a negative voltage supply is given to P-region and a positive supply is given to N-region, then the flow of current will be changed from the prior case.

The movement of electrons flow of current can be easily understood by following the energy band diagram for this case.

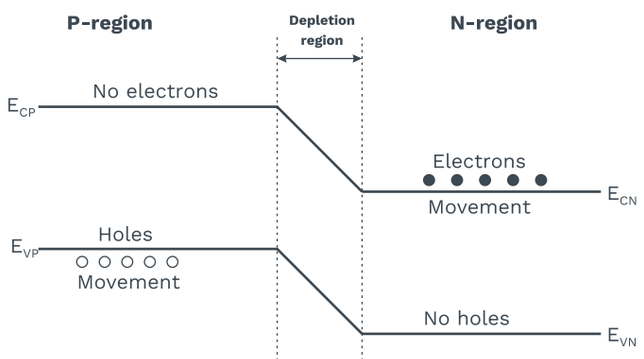


Fig. 2.32 PN Junction Diode Before External Supply

- After external supply

Previous Years' Question

At 300 K, for a diode current, if 1 mA, a certain germanium diode requires at the forward bias of 0.1435 V, whereas certain silicon diode requires at the forward bias of 0.718 V. Under the conditions stated above, the closest approximation of the ratio of reverse saturation current in germanium diode to that in silicon diode is:

a) 1 **b)** 5
c) 4×10^3 **d)** 8×10^3

Rack Your Brain

Do all special diodes require heavy doping?

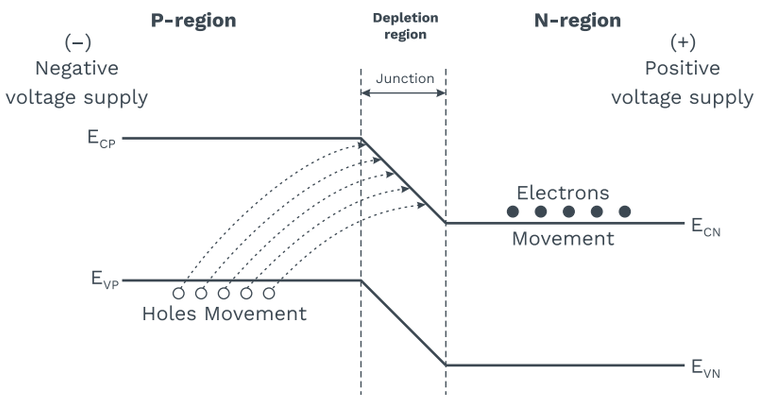


Fig. 2.33 PN Junction Diode After External Supply

Since P-region has no electrons, there will be no flow of current.
The V–I characteristic graph can be redrawn according to this arrangement.
The electron's bonds are broken and thus creating a huge number of electrons at a very high voltage.
After V_B (Break down voltage), bonds are broken-down generally.
The breakdown voltage (or Zener voltage) is represented as:

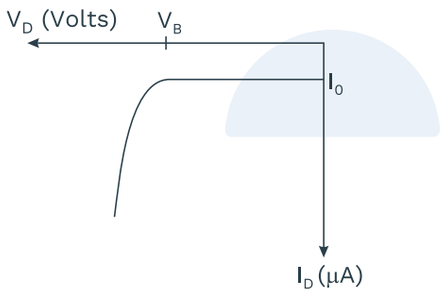


Fig. 2.34 V-I Characteristics of Diode in Breakdown Condition

- Overall,

In forward biasing:
P-region is supplied with a positive voltage, and the N-region is supplied with a negative voltage.
The diode current, $I_D = 0$ A for $V < V_0$ and I_D increases for $V > V_0$.

- For reverse biasing:

P-region is supplied a negative voltage supply, and N-region is given positive supply

$I_D = I_0 \cong 0$ A, for $V < V_B$
and I_D increases for $V > V_B$

Note:
The diode forward junction voltage cannot exceed V_0.
The diode reverse junction voltage cannot exceed V_B (breakdown) voltage.

- Consider a diode given voltage as follows:

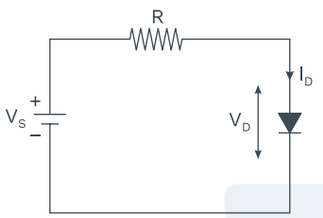


Fig. 2.35 Circuit Containing Diode

Here for this case,
If $V_S > V_0$, then I_D will increase (excess current).
But when $V_D = V_0$, then it is fixed.
The above circuit arrangement depicts the forward bias condition.
In reverse bias,
If $V_S > V_B$, then only diode current (I_D) increases.
But if $V_S < V_B$, then the diode current will be approximately zero or $I_D \cong I_0 = 0A$ and $V_S = V_D$.
The voltage will completely appear across the diode.

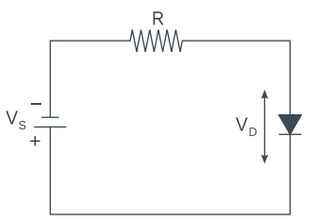


Fig. 2.36 Circuit Containing Diode

The V–I characteristic graph of forward and reverse biasing condition:

Rack Your Brain

How can a diode withstand large power dissipation without getting damaged?

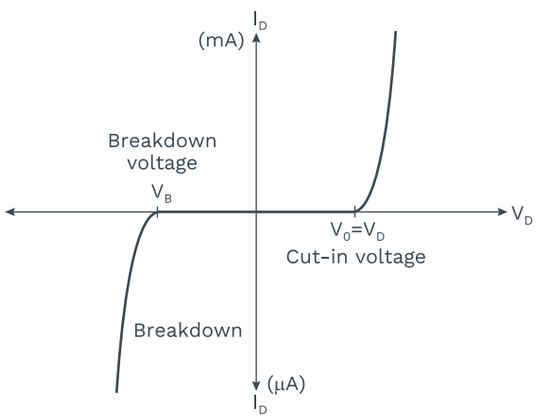


Fig. 2.37 V-I Characteristics of Diode

- At $V_0 = V_D$, as the V_S increases, the diode current I_D also increases. The extra voltage is getting converted into current.
- In a forward-biased condition, the diode looks as follows:

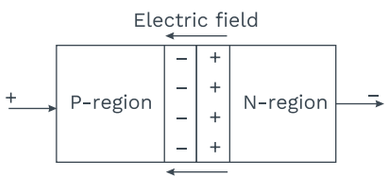


Fig. 2.38 Diode in Forward Bias Condition

When $V < V_0$, there will be an opposing electric field near the junction and the width of the depletion the region is reduced.

Holes for the P-region will try to pass the depletion of the region; the positive charge gets recombined with the negative charge. Therefore, the width of the depletion region reduces.

- Similarly, the electrons on the N-region will cross the depletion region and get recombined with the positive charge present at the N-region near the junction.

There will be no current because the holes on P-region & electrons on N-region are just trying to cross the depletion region.

Previous Years' Question

Consider an abrupt p-n junction. Let V_{bi} is the built-in potential of this junction and V_R be the applied reverse bias. If the junction capacitance (C_j) is 1 pF for $V_{bi} + V_R = 1$ V, then for $V_{bi} + V_R = 4$V, C_j will be:

a) 4 pF **b)** 2 pF
c) 0.25 pF **d)** 0.5 pF

Therefore, $I_D = 0$ A
and $V_D = V$
where

V = External voltage

V_D = Diode voltage

- Due to the above reason, the depletion region gets narrowed.

The forward bias of a diode with a circuit is as follow:

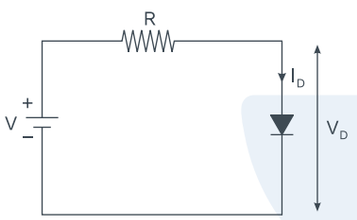


Fig. 2.39 Diode in Forward Bias Condition

If $V < V_0$

Then $I_D = 0$A and $V_D = V$

- Depletion region width (W_N or W_P or W) will be reduced.

$$V_D = \frac{eN_AN_DW^2}{2 \in (N_A + N_D)}$$

where, W = Width of depletion region

- Before cut-in voltage:

When $V < V_0$, when:

$$(V_0 - V) = \frac{eN_AN_D}{2 \in (N_A + N_D)} \cdot x^2$$

where

x = Width of the depletion region is reduced from W or reduced width at V voltage.

For $V_0 = V$ the x = 0 width of depletion region vanishes.

For $V > V_0$

Then,

Previous Years' Question

In an abrupt p-n junction, the doping concentrations on the p-side and n-side are $N_A = 9 \times 10^{16}/cm^3$ and $N_D = 1 \times 10^{16}/cm^3$, respectively. The p-n junction is reverse biased, and the total depletion width is 3 μm. The depletion width on the p-side is:

a) 2.7 μm **b)** 0.3 μm
c) 2.25 μm **d)** 0.75 μm

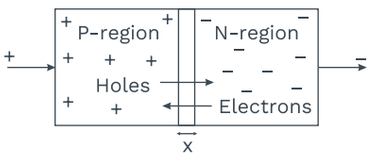


Fig. 2.40 Diode in Forward Bias Condition

- There will be a flow of current, $I_D \neq 0$

There will be a free flow of majority carriers, or it can be said the majority of the carrier are drifting.

Since the holes will move to N-region, they will not recombine with minority carriers immediately.

- The recombination will occur gradually, and ultimately, the strength will get back to a steady-state value.
- This drifting of free electrons will cause diffusion since the majority carriers become minority carriers.

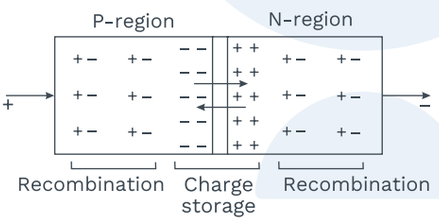


Fig. 2.41 Recombination of Charge Carriers Under Forward Bias Condition

This charge storage is usually dynamically updated due to new electrons or holes.

- When a forward bias is applied, the potential is greater than the cut-in voltage, the majority carrier freely cross-overs the junctions.
- After they freely pass the depletion region, it generally does recombinations fastly or minority carrier injection at one end to the place, which diffuses and moves forward.
- These minority charges diffuse gradually and then form the other end.

This is how continuity of flow works, and it makes a closed circuit.

Thus, the diode current depends on the diffusion process. Faster the diffusion process, the larger the current.

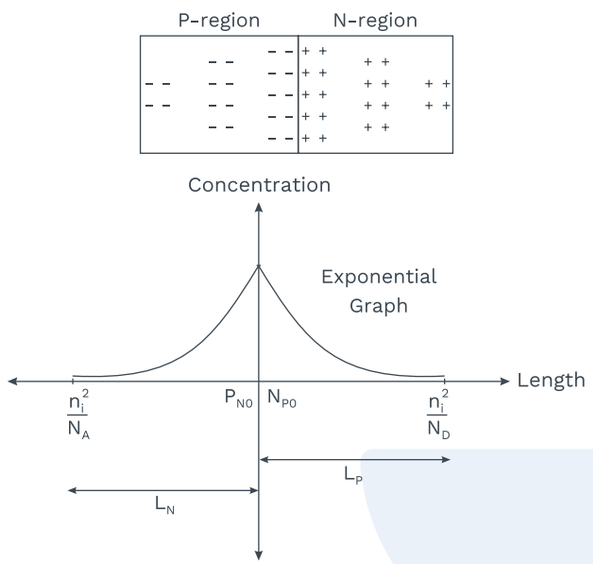


Fig. 2.42 Carrier Concentration in Diode

Rack Your Brain

How much is the breakdown voltage for an ordinary diode?

L_P and L_N are mean diffusion lengths in the P and N regions.

$\frac{n_i^2}{N_A}$ and $\frac{n_i^2}{N_D}$ are steady-state concentration values.

- The rate of accumulation of minority carriers depends on the voltage supply-

  P_{N0}= Concentration of holes (minority carriers) reaching the N-region.

  N_{P0}= Concentration of electrons (minority carriers) reaching the P-region.

- The current will depend on diffusion in both regions.

In other words, diode continuity or current solely depends on the diffusion of the minority rather than the free movements of majorities.

- It is concluded that:

The current travels from the P-region to the N-region in forward bias, and the movement journey of the electron starts from E_{CN} (conduction band energy level in N-region) and goes to the E_{VP} (valence band energy level in P-region).

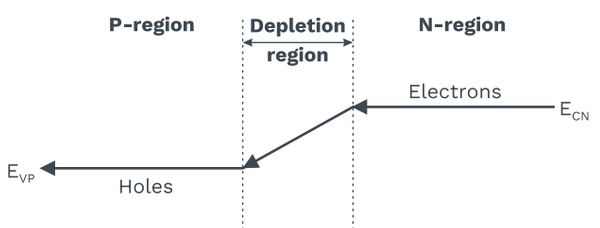


Fig. 2.43 Movement of Electrons in Diode

- It is important to note that charges moving in N-region and E_{CN} are due to electrons, but the movement of charges in E_{VP} in P-regions are due to holes.
- Diode current equations:

Consider the diffusion of the holes in the N-region to be one component of the current.

The diffusion current is given by:

$$I_{Diffusion} = eA\,D_P \frac{dP}{dx}$$

where A = Area of the cross-section.

Since we know the cut-in voltage expression is expressed as:

$$V_0 = -V_T \ln\left(\frac{P_P}{P_N}\right)$$

$$\text{or} \quad V_0 = V_T \ln\left(\frac{N_N}{N_P}\right)$$

P_P = Majority carriers on P-region
N_P = Minority carriers on P-region
P_N = Minority carriers on N-region
N_N = Majority carriers on N-region

Grey Matter Alert!

The laboratory diode 1N4007 has a breakdown voltage of 400 volts and a cut-in voltage of 0.7 volts.

- If $V > V_0$, then the excess minority carriers injected on the other side will rise from its steady-state value N_{P0} or P_{N0} to N_P or P_N, respectively.
- If we apply any voltage V, then the minority carrier concentration increases very rapidly due to the injection. i.e., the holes P_N will rise exponentially, as:

$$P_N = P_{N0}\, e^{\frac{V}{V_T}}$$

where,

P_N is injected concentration value.

$$P_{N0} = \frac{n_i^2}{N_A} \text{ (Steady-state value)}$$

V = Applied potential

Therefore, the

$$\frac{dP}{dx} = \frac{\text{Change from } P_N \text{ to } P_{N0}}{\text{Length between } P_N \text{ \& } P_{N0}}$$

$$\frac{dP}{dx} = \frac{P_N - P_{N0}}{L_P}$$

The diffusion current (I_D) (or diode current) due to holes on P-region is:

$$I_{DP} = e A D_P \left(\frac{P_{N0}\, e^{\frac{V}{V_T}} - P_{N0}}{L_P} \right)$$

$$\text{or} \quad I_{DP} = \frac{e A D_P P_{N0}}{L_P} \left(e^{\frac{V}{V_T}} - 1 \right)$$

Similarly, on the N-region due to electrons:

$$I_{DN} = \frac{e A D_N N_{P0}}{L_N} \left(e^{\frac{V}{V_T}} - 1 \right)$$

The total diode current (I) will be:

$$I = I_{DP} + I_{DN}$$

$$\therefore \quad I = e A \left(\frac{D_P \cdot P_{N0}}{L_P} + \frac{D_N \cdot N_{P0}}{L_N} \right) \left(e^{\frac{V}{V_T}} - 1 \right)$$

$$\text{Let } e A \left(\frac{D_P \cdot P_{N0}}{L_P} + \frac{D_N \cdot N_{P0}}{L_N} \right) = I_0$$

$$I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right)$$

If V is a positive value and $V > V_0$, then

$I = I_0 = e^{V/0.025} = I_0\, e^{V40}$

where I_0 is in μA & if we take V = 1V or less then 1.

Then, I will be in mA.

Also, if $V > V_0$, then I_D increases in mA where $V_D \cong V_0$

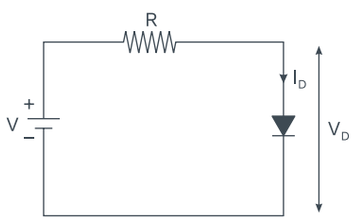


Fig. 2.44 Circuit Having Diode in Forward Bias

The excess of V is applied, then V_D is converted into current and diode voltage will remain as $V_D \cong V_0$.

Rack Your Brain

What should be done to have more power dissipation?

Thus, $V_{increases}$, I_D also increases and $V_D \cong V_0$
In forward bias:

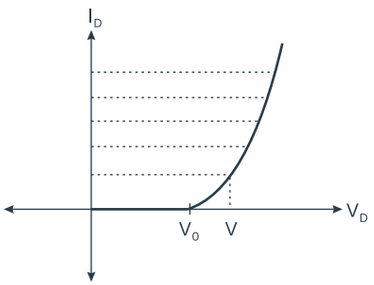


Fig. 2.45 V-I Characteristics of Diode in Forward Bias

From the above figure, we can conclude that whenever the power supply voltage exceeds the cut-in voltage, the current increases very less with voltage.

Source voltage (V_s)	Diode voltage (V_D)	Diode current (I_D)	Voltage excess current (V_R)
0.2	0.2	0	0
0.7	0.7	0	0
0.9	0.7	0.2/R	0.2
1.5	0.7	0.8/R	0.8
2.0	0.7	1.3/R	1.3

Table 2.1

The above table is just an example to show that if $V > V_D$. Then the excess current will come across resistance (R), and the diode current will be varied accordingly.
The diode resistance non-ideal behaviour of the diode is here neglected during graph analysis.

- The approximate or linear model of a diode (for forward biasing conditions) is shown below:

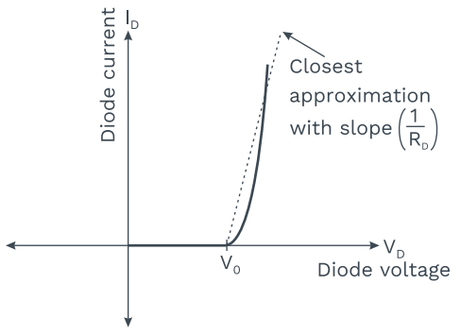


Fig. 2.46 Approximate V-I Characteristics of Diode in Forward Bias

R_D = Diode resistance or average resistance
V_0 = Cut-in voltage
Hence, it is forward bias.

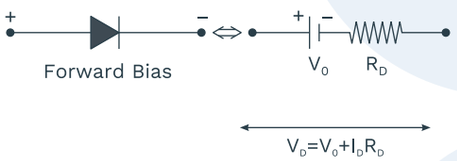


Fig. 2.47 Diode with Cut-in Voltage and Internal Resistance

For a small value of I_D, $V_D = V_0$.

- And if R_D is negligible, then also the diode is regarded as a simple voltage source.

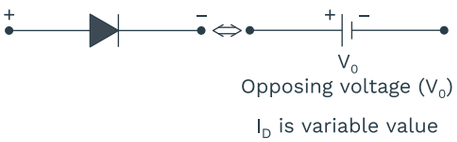


Fig. 2.48 Diode Having Cut-in Voltage

If V_0 is ignored/negligible, the diode will be a short circuit.

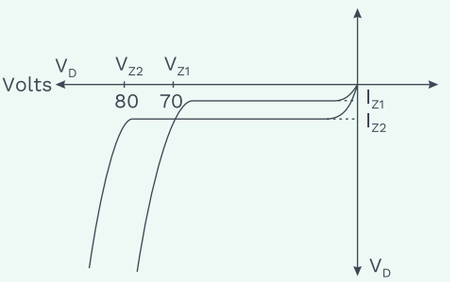


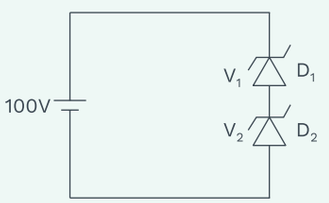
Fig. 2.49 Diode with No Cut-in Voltage

- Consider the following circuit.

Previous Years' Question

A V-I characteristic graph is given for a circuit.





Which of the following is correct?

a) $V_1 = 70V$; $V_2 = 30V$ as $V_{Z1} < V_{Z2}$
b) $V_1 = 20V$; $V_2 = 80V$ as $V_{Z1} < V_{Z2}$
c) $V_1 = 70V$; $V_2 = 30V$ as $V_{01} < V_{02}$
d) $V_1 = 20V$; $V_2 = 80V$ as $V_{01} < V_{02}$

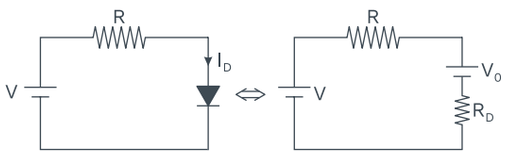


Fig. 2.50 Circuit Containing Diode with Internal Resistance

The diode current will be:

$$I_D = \frac{V - V_0}{R + R_D}$$

On ignoring R_D or R_D = 0 Ω

Then $I_D = \frac{V - V_0}{R}$

If V_0 = 0V and R_0 = 0 Ω

Then, $I_D = \frac{V}{R}$ (Short-circuit)

Example 7:
Consider a circuit with an ideal diode placed across a voltage source with $V > V_0$, then the role of resistance in the circuit is:

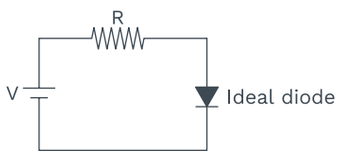


Fig. 2.51 Circuit Containing Ideal Diode

a) It protects the diode from heavy currents.

b) It protects the power supply from heavy currents.

c) It protects the diode from heavy voltage.

d) It is just a switching load in the circuit.

Sol:
In an ideal diode for $V > V_0$, the diode becomes short circuit, and without R, the entire power supply is short-circuited.

Rack Your Brain

What is the regulator voltage across the load?



Fig. 2.52 Ideal Diode in Forward Bias Condition

When V is increasing, when if it is a non-ideal diode or ideal diode.
And $V > V_0$, then $(V–V_0)$ will appear across the resistance.

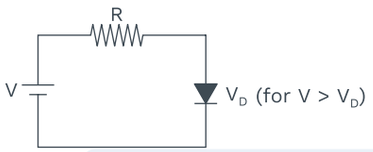


Fig. 2.53 Practical Diode in Forward Bias

The series resistance accepts the extra voltage which the diode cannot support.

i.e. $(V – V_0)$ appears across R.

∴ The resistance in the circuit protects the power supply from the heavy current.

Hence, the correct option is b).

- As we know:

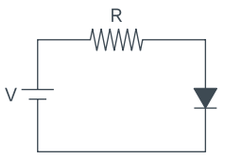


Fig. 2.54 Practical Diode with Internal Resistance

$$I_D = \frac{V - V_D}{R + R_D}$$

The power dissipation will be:

$P = V_0 I_D$ (mW)

- If I_D is increased rapidly since V is increasing and exceeds the power rating (mW) then the diode will be damaged after a certain mW value.

Previous Years' Question

A silicon p-n junction at a temperature of 20°C has a reverse saturation current of 10 pico-Amperes (pA). The reverse saturation current at 40°C for the same bias is approximately:

a) 30 pA **b)** 40 pA
c) 50 pA **d)** 60 pA

REVERSE-BIAS OPERATION

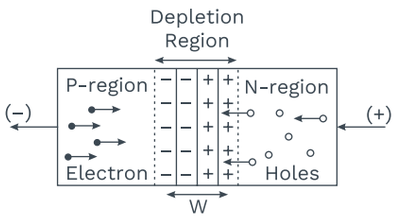


Fig. 2.55 Diode in Reverse Bias Condition

- The reverse bias operation can be understood if we connect N-region to the positive supply and the P-region with the negative supply.
- The holes (minority carriers) in N-region move from N-region towards the junction and increase the ion concentration around the depletion region.
- So, ultimately, the width of the depletion region increases.
- Reverse bias increases the thickness potential barrier, because more and more are coming to junctions.

The cut-in voltage is given as:

$$V_0 = \frac{e N_D N_A}{2 \in (N_D + N_A)} W^2$$

If $V < V_0$, we know that:
For any negative voltage,

$$V_0 - V = \frac{e N_D N_A \, W^2}{2 \in (N_D + N_A)}$$

Here, V has a negative value & hence, the width will be increased.

$$\therefore \quad V_0 - (-V) = \frac{e N_D N_A}{2 \in (N_D + N_A)} W^2$$

$$\text{or} \quad V_0 + V = \frac{e N_D N_A}{2 \in (N_D + N_A)} W^2$$

- For V as positive, the width will be decreased.
- Also, the minimum current flows as reverse current due to the minority charge carriers, which are fixed and constant current.

Therefore,

$$I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right)$$

Now, If $V < 0$

$$I = I_0\left(e^{(40)(-V)} - 1\right) \text{ or } I = I_0\left(e^{-40V} - 1\right)$$

and if we get V = 1 Volt

Then $\frac{1}{e^{40}} \simeq 0$ or is ignored for most voltages.

$\therefore \quad e^{-40V} \approx 0$

Then $I = I_0 (0 - 1) = -I_0$ for $V < 0$.

Now, this I_0 is referred to as reverse saturation current.

Also, in forward bias, we know that,

$$I = eA D_P \frac{dP}{dx} = eA D_P \left(\frac{P_N - P_{N0}}{L_P}\right)$$

where P_N = Injected holes concentration in forward bias.

P_{N0} & also $P_N = P_{N0}\, e^{\frac{V}{V_T}}$

$\therefore \quad I = I_0\left(e^{\frac{V}{V_T}} - 1\right)$

$I = K\, e^V$ and here, V is positive and therefore $\left(I_0\, e^{\frac{V}{V_T}}\right)$ is a large value

$$\left(e^{\frac{V}{V_T}} >>> I_0\right)$$

$\therefore \quad e^{\frac{V}{V_T}}$ exists only in forward bias.

- In short, during forward biasing:

$$I = I_0\left(e^{\frac{V}{V_T}} - 1\right)$$

here, $V > V_0$ and V is a positive value.

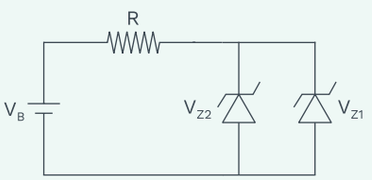
Thus, the equation becomes $I = I_0\, e^{\frac{V}{V_T}}$

- When V = is negative and has any value.

Then $\left(e^{\frac{V}{V_T}}\right)$ doesn't have much role, the equation becomes $I = -I_0$.

Previous Years' Question

Consider the two Zener diodes having the same reverse saturation current but a different breakdown voltage as $V_{Z1} > V_{Z2}$, $V_B > V_{Z1}$.



Which of the two diodes will never be in the breakdown?

This occurs in reverse biasing conditions only.

- The key points of the above discussion are:
  In forward biasing
  $V \cong V_0$ & I_D is variable
  $V = V_0 + I_D R_D$
- If R_D is ignored, then $V \cong V_0$
  In reverse biasing:
  $I = I_0$ & V is variable
- The V – I characteristic for reverse bias:

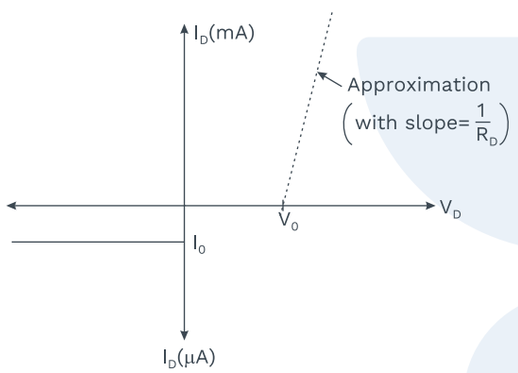


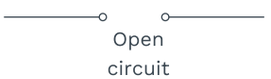
Fig. 2.56 V-I Characteristics Under Reverse Bias Condition

- Approximation model or Linear approximate model.



Fig. 2.57 Approximate Model of Diode in Reverse Bias

- If I_0 = 0A, a diode is considered an open circuit.



(In reverse bias)

Fig. 2.58 Diode as Open-Circuit in Reverse Bias

Previous Years' Question

A silicon p-n junction diode under reverse bias has a depletion region of width 10 μm. The relative the permittivity of silicon ε_r = 11.7 and the permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. The depletion the capacitance of the diode per square meter is:

a) 100 μF/m^2 **b)** 10 μF/m^2
c) 1 μF/m^2 **d)** 20 μF/m^2

The Dynamic resistance:

In piecewise linear models, there is static resistance (R_D) (where the slope of current = $\frac{1}{R_D}$ = Fixed).

Practically, dynamic resistance is represented as:

$$r_D = \frac{dV}{dI}$$

The dynamic resistance can be derived as:

$$I = I_0\left(e^{\frac{V}{V_T}} - 1\right) \text{ and } \frac{dI}{dV} = I_0\left(e^{\frac{V}{V_T}}\right)\frac{1}{V_T}$$

Since, $I_0 = \left(e^{\frac{V}{V_T}}\right) \approx I$

$$\therefore \quad \frac{I}{V_T} = \frac{1}{r_D}$$

The dynamic resistance (r_D) will be:

$$r_D = \frac{V_T}{I}$$

The slope of the linear approximation model is static resistance i.e. $\frac{1}{R_D}$

Note:

The other ways to define I_0 are:

$$I_0 = e\,A\left(\frac{D_P \cdot P_{N0}}{L_P} + \frac{D_N \cdot P_{P0}}{L_N}\right)$$

$$I_0 = e\,A\left(\frac{D_P \cdot n_i^2}{N_D \cdot L_P} + \frac{D_N \cdot n_i^2}{N_A L_N}\right)$$

or $$I_0 = e\,A\left(\frac{D_P}{N_D L_P} + \frac{D_N}{N_A L_N}\right)$$

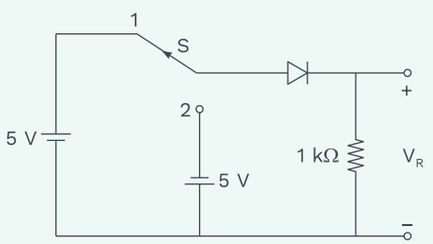
It is also said that I_0 is highly (n_i^2) dependent and the n_i^2 is (T^3) temperature-dependent.

Also, we know that:

$$D_P = \frac{L_P^2}{\tau_P}$$

Previous Years' Question

In the circuit shown below, the switch was connected to a position 1 at t < 0 and at t = 0, it was changed to a position 2. Assume that the diode has zero voltage drop and a storage time t_S for $0 < t \le t_S$, V_R is given by (all in Volts)



a) $V_R = -5$
b) $V_R = +5$
c) $0 \le V_R < 5$
d) $-5 \le V_R < 0$

where L_P = mean diffusion length
τ_P = lifetime

Now, substituting in the above equation:

$$I_0 = e A n_i^2 \left(\frac{L_P}{N_D \cdot \tau_P} + \frac{L_N}{N_A \cdot \tau_N} \right)$$

Example 8:

The value of I_0 is very large for:

a) Silicon
b) Germanium
c) Equal for both silicon and germanium
d) Data insufficient

Sol:

Since we know that,

$n_i = 1.5 \times 10^{10}$ for silicon and $n_i = 5 \times 10^{13}$ for germanium. As n_i is very large

$\therefore \quad I_{0(Ge)} >> I_{0(Si)}$

Hence, the correct option is b).

Note:

If the value of I_0 increases as the n_i increases. Since $T^{3/2}$ increases, it is concluded that the I_0 is temperature-sensitive for every 1°C, the value of I_0D increases by 7% (for silicon) & 11% (for germanium) will be:

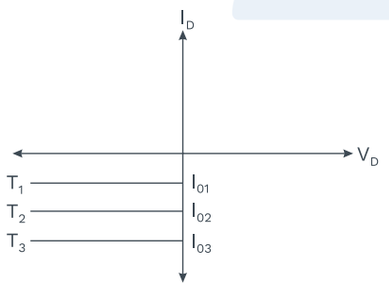


Fig. 2.59 V-I Characteristics Under Reverse Bias Condition at Different Temperature

- It is concluded that for a 1°C increase in silicon, the I_0 will increase by (1.07) I_0.
- For 2°C increase, I_0 will be $(1.07)I_0 \times 1.07$.

But after 10°C, its $(1.07)^{10}$ I_0 increase.

- Thus, for every 10°C, the I_0 increase by 100%, i.e., it becomes twice (double).
- For 20°C, it will be four times & thus goes on.

$\therefore$ The current I_0 for any temperature T_2 will be:

$$I_{02} = I_{01} \left(2^{T_2 - T_1/10} \right)$$

Rack Your Brain

Why is the negative resistance region important?

where I_{01} = Reverse saturation current at temperature T_2.
And the above formula is for germanium.

Example 9:
A P-N junction is forward biased; the sequence of events given below describes the mechanism of current flow is:

a) Injection, subsequent diffusion and recombination of minority carriers.
b) Injection, subsequent drift, and then generation of minority carriers.
c) Extraction, subsequent diffusion, and generation of minority carriers.
d) Extraction, subsequent drift and combination of minority carriers.

Sol:
In forward biasing, to produce the movements of electrons, the injection is done from P to N-region or N to P-region.
Then the subsequent diffusion process occurs along with the recombination as the next process for minority carriers.
Then, the correct option is a).

- When the flow is heavy, then there will be no accumulation, and the current will increase with $V = V_0$ during forward bias.

In forward bias:
If forward bias voltage is increased, then junction voltage will be constant, but the device current will increase.

In reverse bias:
When the flow of current is not possible, the accumulation will be there.
$I = I_0$ and bias voltage will increase & junction voltage will also increase.

Previous Years' Question

Which of the following is not associated with a p-n junction?
a) Junction capacitance
b) Charge storage Capacitance
c) Depletion capacitance
d) Channel length Modulation

Example 10:
The diode has V_{ON} = 0.7 Volts but is ideal in nature. Find I_2 (in mA).

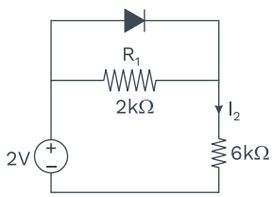


Fig. 2.60 Circuit Containing Diode

Sol:
Assume the diode is in forward bias and operates beyond the cut-in voltage.
The voltage across R_1 & diode will be 0.7V.

$V_{R1} = V_D = 0.7$ volts
$V_{R2} = 1.3$ volts

∴ Current through $R_1(I_{R1})$, i.e. IR_1 will be:

$$I_{R1} = \frac{0.7}{2\,k\Omega} = 0.35\,mA$$

$$I_{R2} = \frac{1.3}{6\,k\Omega} = 0.21mA$$

This is not possible.

Now, consider the diode to be forward-biased & voltage lesser than cut-in voltage, i.e. $V < V_0$.

Then I_0 = 0A (open circuit)

The equivalent circuit will be:

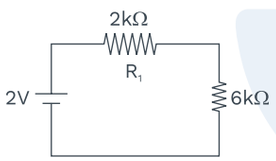


Fig. 2.61 Circuit when $V < V_0$

∴ Current will be:

$$I = \frac{2}{8\,k\Omega} = 0.25\,mA$$

Hence, the correct answer is 0.25 mA.

Example 11:

For the given circuit:

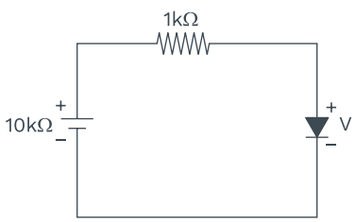


Fig. 2.62 Circuit Containing Diode

The I-V characteristic of the diode is given as:

$$I = \begin{cases} \frac{V - 0.7}{500}\,A & , \; V \geq 0.7\,Volts \\ 0A & , \; V < 0.7\,Volts \end{cases}$$

The current flowing in the above circuit is:

a) 6.2 mA **b)** 6.2 A **c)** 10 mA **d)** 10.5 mA

Previous Years' Question

Compared to p-n junction with $N_A = N_D = 10^{14}/cm^3$, which one of the following statement is TRUE for a p-n junction with $N_A = N_D = 10^{20}/cm^3$?

a) Reverse breakdown voltage is lower, and depletion capacitance is lower.

b) Reverse breakdown voltage is higher, and depletion capacitance is lower.

c) Reverse breakdown voltage is lower, and depletion capacitance is higher.

d) Reverse breakdown voltage is higher, and depletion capacitance is higher.

Sol:

The given V-I relationship for the diode in the circuit is:

$$I = \begin{cases} \frac{V - 0.7}{500} A & , \ V \geq 0.7 \text{ Volts} \\ 0A & , \ V < 0.7 \text{ Volts} \end{cases}$$

But we know that:

$$I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right) \qquad \{\text{Linear model}\}$$

From the above-given data, we can conclude that:

$$I = \frac{V - 0.7}{500}$$

Voltage across the diode:

-and $V = 0.7 + 500\ I = V_D$

$R = 500\ \Omega$

$V = 0.7$ Volts

Now, redrawing the given circuit,

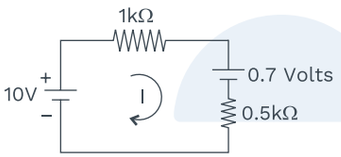


Fig. 2.63 Diode Replaced by a Voltage Source and Internal Resistance

∴ The current will be:

$$I = \frac{10 - 0.7}{1 + 0.5} \text{ mA}$$

$I = 6.2$ mA

Hence, the correct option is a).

Rack Your Brain

Which of the following are active devices or passive devices? Zener diode, Ordinary diode, Schottky diode?

Example 12:

For a silicon diode with long P and N regions, the acceptor and the donor concentrations are $10^{17}/cm^3$ and $10^{15}/cm^3$, respectively.

The lifetimes of holes in the P-region and electrons in the N-region are 100 μs. The hole and electron diffusion coefficients are 36 cm^2s^{-1} and 49 cm^2s^{-1}, respectively. Assume that a forward voltage of 208 m Volts is applied across the diode, then the hole current density injected from P-region to N-region is:

a) 0.5147 nA cm^{-2} **b)** 0.4145 nA cm^{-2}

c) 0.3142 nA cm^{-2} **d)** 0.2141 nA cm^{-2}

Sol:
Given:
Acceptor concentration = $10^{17}/cm^3$
Donor concentration = $10^{15}/cm^3$
Lifetimes of electrons and holes in N-region and P-region = 100 μs
V_T = 26 mV
The hole diffusion coefficient = 36 cm^2s^{-1}
Electron diffusion coefficient = 49 cm^2s^{-1}
And we know that:

$$I_{Holes} = eA D_P \cdot \frac{dP}{Dx}$$

$$I_{Holes} = eA D_P \cdot \left(\frac{P_{N0} \cdot e^{\frac{V}{V_T}} - P_{N0}}{L_P} \right)$$

$$I_{Holes} = \frac{eA D_P}{L_P} P_{N0} \left(e^{V/V_T} - 1 \right)$$

The current density is:

$$J = \frac{I_{Holes}}{A} = \frac{e D_P}{L_P} \frac{n_i^2}{N_D} \left(e^{V/V_T} - 1 \right)$$

And we know that:

$$D_P = \frac{L_P^2}{\tau_P}$$

$$= 0.5147 \text{ nA cm}^{-2}$$

Hence, the correct option is a).

Example 13:
A P-N step junction diode with a contact potential of 0.65 Volts has a depletion width of 1 μm at equilibrium. The forward voltage (in Volts) at which this width reduces to 0.6 μm is _____. (up to three decimal places)

Sol:
Given:
Contact potential = 0.65 Volts
Depletion width (w) = 1 μm = 1×10^{-6} m
We know that the cut-in voltage is:

Previous Years' Question

For a silicon diode with long p and n regions, the acceptor and donor impurity concentrations are 1×10^{17} cm^{-3} and 1×10^{15} cm^{-3}, respectively. The lifetime of electrons in the p-region and holes in n region are both 100 μσ. The electrons and hole diffusion coefficients are 49 cm^2/s and 36 cm^2/s respectively. Assume kT/q = 26 mV, the intrinsic carrier concentration is 1×10^{10} cm^{-3} and q = 1.6×10^{-19} C. When a forward voltage of 208 mV is applied across the diode, the hole current density (in nA/cm^2) injected from the p-region to n-region is _____.

$$V_0 = \frac{e N_D N_A \cdot w^2}{2 \in (N_A + N_D)}$$

For forward voltage:

$$(V_0 - V) = \frac{e N_D N_A \cdot w^2}{2 \in (N_A + N_D)} \qquad \{V < V_0\}$$

For $W_0 = 1\ \mu m$ at $V_0 = 0.6V$

For $W_v = 0.6\ \mu m$ at forward voltage (V)

$$\therefore \quad \frac{V_0}{V_0 - V} = \left(\frac{W_0}{W_v}\right)^2$$

$$\frac{0.65}{0.65 - V} = \left(\frac{1}{0.6}\right)^2$$

$$\frac{0.65}{0.65 - V} = \frac{1}{0.36}$$

$V = 0.65 - 0.234$

$V = 0.416$ Volts

- In forward biasing:

If $V_s > 2V_0$, where V_s = source voltage

$$\left.\begin{aligned} V_{D1} &= V_0 \\ V_{D2} &= V_0 \end{aligned}\right\} \begin{array}{l}\text{for Single or double} \\ \text{Diode combination}\end{array}$$

and series connections, $V_D = V_0$.

- In reverse biasing:

Single diode = $V_s = V_D$

Double diode (in series) combination:

$$V_{D2} = \frac{V_s}{2}$$

Note:

There is no extra advantage at forward biasing,

Since, $V_D = V_0$

But it varies in the case of reverse biasing.

Previous Years' Question

The doping concentration on the p-side and n-side of a silicon diode is $1 \times 10^{16}\ cm^{-3}$ and $1 \times 10^{17}\ cm^{-3}$, respectively. A forward bias of 0.3 V is applied to the diode. At T = 300 K, the intrinsic carrier concentration of silicon $n_i = 1.5 \times 10^{10}\ cm^{-3}$ and $\frac{kT}{q} = 26\,mV$. The electron concentration at the edge of the depletion region on the p-side is:

a) $2.3 \times 10^{9}\ cm^{-3}$

b) $1 \times 10^{16}\ cm^{-3}$

c) $1 \times 10^{17}\ cm^{-3}$

d) $2.25 \times 10^{6}\ cm^{-3}$

Example 14:
Consider the given circuit:

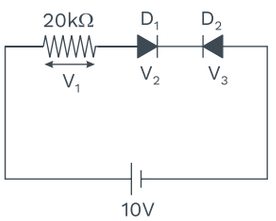


Fig. 2.64 Circuit Containing Two Diodes

Assume that the reverse current in D_2 is twice the reverse current in the D_1 diode.

Evaluate V_1, V_2 and V_3. Here, we assume that mobility does not get affected due to doping.

Sol:
From the given circuit, it is clear that biasing is reverse biasing.

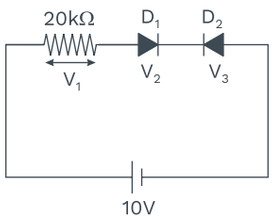


Fig. 2.65 Circuit Containing Two Diodes

The voltage across 20kΩ

$V_1 = (20\ I_{02})K$ Volts

and it is given that:

$I_{02} = 2I_{01}$

Across diode–2 (D_2), the current is:

$$I_{02} = I_{02}\left(e^{V_3/\eta V_T} - 1\right)$$

And $V_3 = 0$ V
or $V_3 = -\infty$
The V_2 voltage will be across D_1:

$$I_{02} = I_{01}\left(e^{V_2/\eta V_T} - 1\right)$$

$$2I_{01} = I_{01}\left(e^{V_2/\eta V_T} - 1\right)$$

Previous Years' Question

A silicon p-n junction is forward biased with a constant current at room temperature. When the temperature is increased by 10°C, the forward bias voltage across the p-n junction:
a) Increases by 60 mV
b) Decreases by 60 mV
c) Increases by 25 mV
d) Decreases by 25 mV

$\therefore \quad V_2 = \eta V_T \ln(3)$ volts.

Therefore, $V_1 = (20\ I_{02})$ Volts

$V_2 = \eta V_T \ln(3)$

$V_3 = (10 - V_1 - V_2)$ Volts

Diffusion capacitance and transition capacitance:

- The diffusion capacitance (C_D) is often experienced in forward biasing.

Whereas the transition capacitance (C_T) occurs in reverse biasing.

Diode as a capacitor:

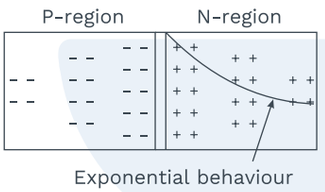


Fig. 2.66 Diode as a Capacitor

In N-region, the holes (minority charge carriers) obey exponential function behaviour, and similar behaviour is experienced in P-region due to electrons.

- The cross-over carrier and their recombination rate are used to determine the current in the diode.

A high accumulations rate results in less recombination rate & hence they oppose any incoming charge carriers and will not allow any junction.
This is referred to as the charge control description of the diode.
So, the current is ultimately dependent on the diffusion of these carriers rather than the drift of the majority of carriers.

- These accumulated carriers give rise to capacitive effects & which are defined as diffusion capacitance.

Mathematically:

$$C_D = \frac{dQ}{dV} = \frac{dQ}{dt \cdot \frac{dV}{dt}}$$

$$C_D = \frac{dI}{dV} \cdot dt$$

$\tau = dt$ is the time for which the charge is changing.
It is known as the mean lifetime or recombination time of holes/electrons.

$$\therefore \quad C_D = \frac{\tau}{r_D}$$

where r_D = Diode resistance

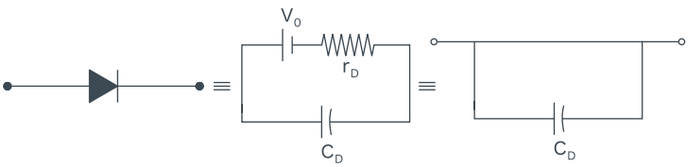


Fig. 2.67 Equivalent Circuit of Diode

Here, r_D is negligible, and V_0 is also ignored.

- **During reverse biasing:**

The capacitive effect of the diode is a time constant (RC) and series resistance.
Hence, R is negligible; thus, the capacitive effect is zero in a diode.
Also, the depletion width (W) = 0

$$C_D = \frac{\in A}{W}$$

$\because$ $W \to 0$, then C_D will be undefined.

$\therefore$ C_D is a large value but has limited application as it is in forward bias.

In reverse biasing:

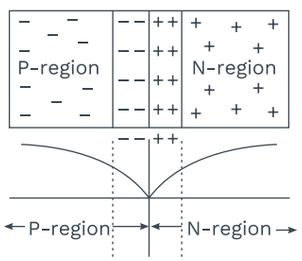


Fig. 2.68 Carrier Concentration in Reverse Bias

- The graph of minority carrier concentration is shown in the above figure as it is clearly seen that it is dropping as it goes near the junction.
- The current flowing from N-region to the P-region is of free velocity is mainly free current, no resistance & it is followed by the drift current (due to electric field).

Previous Years' Question

The donor and accepter impurities in an abrupt junction silicon diode are 1×10^{16} cm^{-3} and 5×10^{18} cm^{-3}, respectively. Assume that the intrinsic carrier concentration in silicon $n_i = 1.5 \times 10^{10}$ cm^{-3} at 300K, $\frac{kT}{q} = 26\,mV$ and the permittivity of silicon $\varepsilon_{Si} = 1.04 \times 10^{-12}$ F/cm. The built-in potential and the depletion width of the diode under thermal equilibrium conditions, respectively, are:

a) 0.7 V and 1×10^{-4} cm
b) 0.86 V and 1×10^{-4} cm
c) 0.7 V and 3.3×10^{-5} cm
d) 0.86 V and 3.3×10^{-5} cm

Hence, the forward current is diffusion-based, whereas the reverse current is drift based.

Minority carriers crossing over the junction generally results in a drop in majority carrier concentration at the junction.

- The current will have a drift component inside the depletion region & it is constantly saturated because the number of carriers is fixed & it doesn't increase their voltage.

Thus, it is referred to as transition capacitance (C_T), which is given as:

$$C_T = \frac{\in A}{W}$$

where W = width of depletion region and W changes as soon as we apply external voltage.

- It can be said that the depletion region is working as a dielectric. (A dielectric consisting of ions and electric field)
- Simple capacitance:

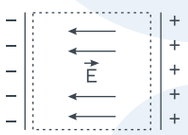


Fig. 2.69 Capacitance in a Capacitor

Diode capacitance:

The transition capacitance is often used in IC fabrication & in most embedded designs.

- In semiconductor technology, silicon materials are used to manufacture the capacitor.
- The transition capacitance is also employed in VCOs and frequency modulator circuits.

Most diodes used transition diodes (C_T) and are also called varicaps or varactor diodes.

$$(V_0 - V) = \frac{eN_AN_D \cdot W^2}{2 \in (N_A + N_D)}$$

$$\text{and } C_T = \frac{\in A}{W}$$

Substituting the values, we get:

$$C_T \propto \frac{1}{\sqrt{V_0 - V}}$$

- **Junction switching times:**

Every capacitive effect results in delays due to charging and discharging,
i.e., $\tau = RC$
The time constant (τ) causes delay. The diffusion capacitance (C_D) causes large delays and is explained as follows:
Forward recovery time due to C_D:

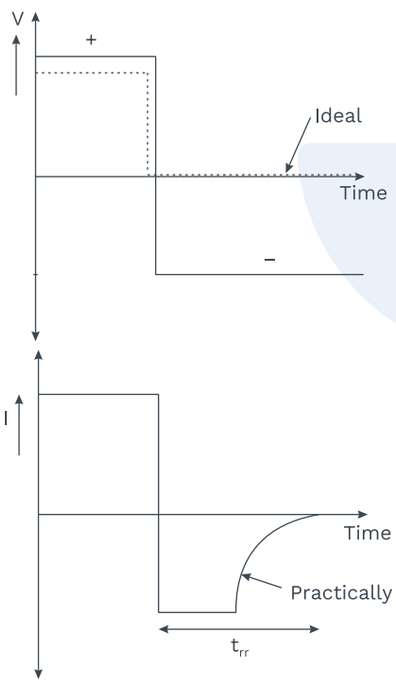


Fig. 2.70 Recovery Time

REVERSE RECOVERY TIME

- When the diode is ideal, the current will drop to zero. But in practical scenarios, there is a reverse current for some time, and then it gradually drops to zero.
- The time duration is referred to as reverse recovery time.
- When a diode is reverse biased suddenly from a forward biased, there is a finite duration of conduction or where the current flows even in the reverse bias.

Previous Years' Question

The depletion capacitance C_j of an abrupt p-n junction with constant doping on either side varies with reverse bias V_R as:

a) $C_j \propto V_R$

b) $C_j \propto V_R^{-1}$

c) $C_j \propto V_R^{-1/2}$

d) $C_j \propto V_R^{-1/3}$

- And the time in which there is conduction, even in reverse biasing, is called reverse recovery time.

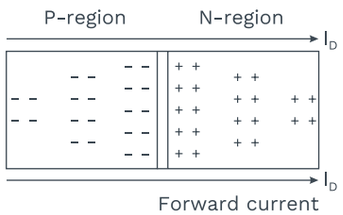


Fig. 2.71 Charge Distribution in Forward Biasing

- From the above figure, it is noted that the charges are accumulated near the junctions in forward biasing.
- But suddenly, if the biasing is changed to reverse biasing, then these accumulated charges go back, and the depletion region gets widened up and reverses current flows.
- These accumulated charges are the main reason for the flow of reverse current or the junction accumulated carriers cross-overs when sudden biasing from forward bias to reverse changes occurs.
- As long as the carriers are present, this reverse current will flow.

There are 2 stages for recovery:

1) A steady-state flow for a finite duration.

2) Decreasing for the next duration.

Thus,

$\therefore$ Recovery time = Storage time + Transition time

$$t_{rr} = t_s + t_t$$

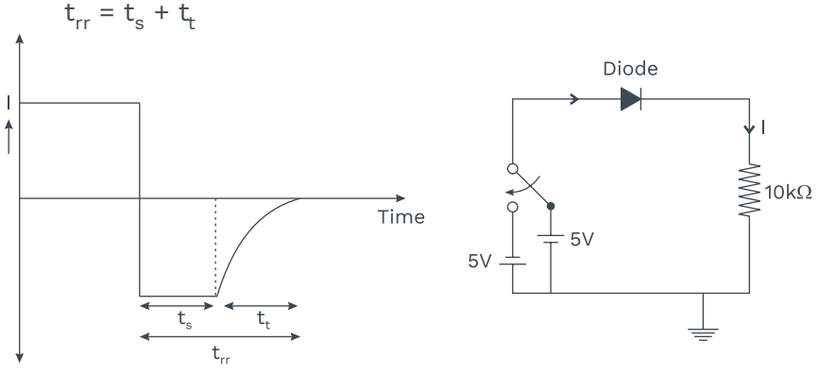


Fig. 2.72 Diode Going from Forward Bias to Reverse Bias

The above-given circuit demonstrates the condition where there is a sudden change from forward biasing to reverse biasing.

- Consider:

If the forward applied voltage (V_F) is (+5V) and reverse applied voltage (V_R) is (–5V),

and $t_s = T_1$, $t_t = T_2$

$\therefore$ $t_{rr} = T_1 + T_2$

if $V_F = 5V$ & $V_R = -10V$

Then, t_s will decrease since larger voltage (thus faster removal of accumulated charges).

$t_t = T_2$ (Same)

$\therefore$ t_{rr} will be reduced overall.

Reverse recovery time limits the diode bandwidth or bit rate or switching speeds.

Note:

The width (W) is directly proportional to the square root of applied voltage (V_a) for only the step-graded diode.

i.e. $W \propto \sqrt{V_a}$

Since, $W = \sqrt{\frac{2 \in (N_A + N_D)}{e N_A N_D}}(V_0 - V)$

For linearly graded,

$W \propto \sqrt[3]{V_a}$

SPECIAL DIODES

- As we know, the power dissipation of a diode is approximately zero in both forward bias and reverse bias.

In forward bias or in (short-circuit ideal diode)

$V_0 I_D \cong 0$

$\because$ $V_0 \cong 0V$

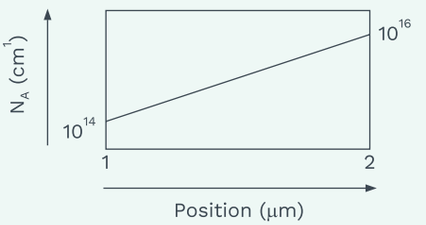
While in reverse bias or open-circuit ideal diode:

$V_0 I_D \cong 0$

$\because$ $V_0 \cong 0V$ & V_D = Variable

Previous Years' Question

The figure below shows the doping distribution in a p-type semiconductor on a log scale.



The magnitude of the electric field (in kV/cm) in the semiconductor due to non-uniform doping is _____.

Previous Years' Question

A p-n step junction diode with a contact potential of 0.65 V has a depletion width of 1 μm at equilibrium. The forward voltage (in volts, correct to two decimal places) at which this width reduces to 0.6 μm is _____.

- Zener diode is a special diode that is mainly designed to operate specifically beyond the large voltage, where the conduction starts due to the breaking of bonds and generation of minority carriers.

This large voltage is termed Zener voltage (V_z).

$\therefore \quad V_z \times I_z = P_z \quad$ {large value}

where V_z = Zener voltage,

I_z = Zener current

- The I-V graph of a normal diode & Zener diode

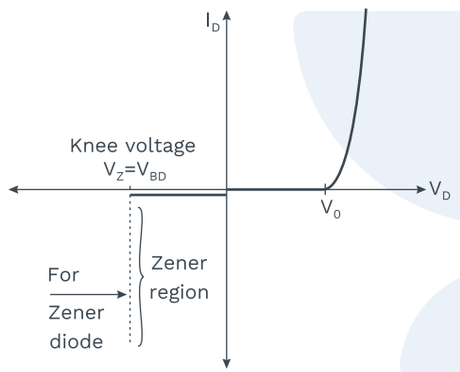


Fig. 2.73 V-I Characteristics of Diode and Zener Diode

The Zener voltage (V_Z) is also called knee voltage.

Features:

- Zener diodes conduct reverse biasing.
- It can withstand large power dissipation.
- The junction voltage is constant for increasing current after conduction.

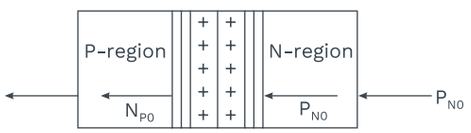


Fig. 2.74 Zener Diode in Reverse Bias

Previous Years' Question

In an ideal p-n junction with an ideality factor of 1 at T = 300K, the magnitude of reverse-bias voltage required to set 75% of its reverse saturation current, rounded off to 2 decimal places, is _____ mV.

[Assume, k = 1.38 × 10^{-23} JK^{-1}, h = 6.625 × 10^{-34} J-s, q=1.602 × 10^{-19} C]

Previous Years' Question

A junction is made between p^- Si with doping density $N_{A_1} = 10^{15}$ cm^{-3} and p Si with doping density $N_{A_2} = 10^{17}$ cm^{-3} cm^{-3}.

Given Boltzmann constant k = 1.38 × 10^{-23} $J\text{-}K^{-1}$, electronic charge q = 1.6 × 10^{-19} C. Assume 100% acceptor ionization. At room temperature (T = 300 K), the magnitude of the built-in potential (in volts, correct to two decimal places) across this junction will be _____.

The reverse bias voltage increase the width of the region & current flow is very low such that the electric field is also increasing (E_{max}) as well as width along and V_D.

$E_{max} \uparrow W \uparrow V_D \uparrow$

We know that:

$$\frac{\partial^2 V}{\partial x^2} = \frac{-eN_D}{\in}$$

$$\frac{\partial V}{\partial x} = E = \frac{-eN_D x_N}{\in} = \frac{-eN_D x_P}{\in}$$

- From this, we get to know that:

Width increases along with E_{max} and V_D.

- If this electric field at this voltage is strong enough to break the bonds, then a huge current will start to flow, and the flow starts the accumulation gets constant.
- At a certain width of a depletion region or a certain diode voltage, the E_{max} at the centre will break the bonds while producing the minority carriers. This is when the current starts to flow.
- The above mechanism is referred to as the Zener mechanism.
- The crossing over of carriers can also break the bonds, provided that the voltage must be large enough.

Thus, at a certain reverse voltage, the minority carriers acquire sufficient energy as they drift in the depletion region and simultaneously initiate the breaking bond on the alter side.

- So, the minority carriers break the bond and this mechanism is referred to as the avalanche breakdown mechanism.

Previous Years' Question

Consider the recombination process via bulk traps in a forward-biased p-n homojunction diode. The maximum recombination rate is U_{max}. If the electron and the hole capture cross-sections are equal, which one of the following is FALSE?

a) U_{max} occurs at the edges of the depletion region in the device

b) U_{max} depends exponentially on the applied bias

c) With all other parameters unchanged, U_{max} increases if the thermal velocity of the carrier increase

d) With all other parameters unchanged, U_{max} decreases if the intrinsic carrier density is reduced

- Since the holes and electrons (minority carriers) in N and P-regions are forcibly crossing the depletion region due to a large voltage applied. This crossing over is also called overcoming the huge potential difference.
- In other words, they are travelling through the huge voltage and acquiring huge energy, in return and initiate the bond-breaking after coming to the other side.

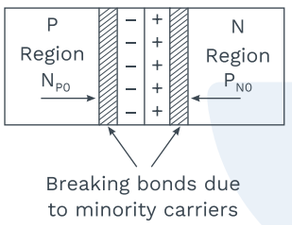


Fig. 2.75 Zener Diode in Reverse Bias

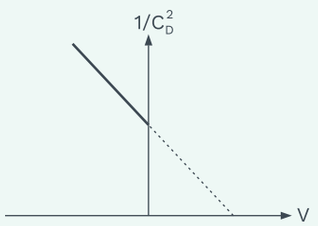
- This reverse current is due to a minority flow current.

If the electric field is breaking the bond to produce a minority carrier, then it is referred to as the Zener breakdown mechanism.

- And if the bonds are breaking due to minority carriers, then this mechanism is termed avalanche breakdown.
- The normal diode will get damaged or burn down at such a large voltage.

Previous Years' Question

A one-sided abrupt p-n junction the diode has a depletion capacitance C_D of 50 pF at a reverse bias of 0.2 V. The plot of $\frac{1}{C_D^2}$ versus the applied V for this diode is a straight line as shown in the figure below. The slope of the plot is _____ $\times 10^{20}$ F^{-2} V^{-1}.



a) – 1.2 **b)** – 5.7
c) – 0.4 **d)** – 3.8

	Zener breakdown	Breakdown avalanche
1.	The breaking of the bond is due to a high electric field in the thin depletion region.	The minority carriers are breaking the bonds.
2.	It occurs in heavily doped diodes.	It occurs in lightly doped diodes.
3.	The thin depletion region exists for the Zener diode.	A thick depletion region and low doping exists in an avalanche.
4.	All the Zener diodes have breakdown voltage that decreases with temperature increases ($T\uparrow\ V_Z\downarrow$).	The breakdown voltage increase with an increase in temperature ($T\uparrow V_B\uparrow$).
5.	In Zener didoes, temperature increases and electric field decreases or lesser Zener voltage will be needed.	In avalanche diodes, temperature increases & reduces the minority carrier's energy due to collision among themselves.
6.	Zener breakdown is observed in Zener diodes, where the breakdown voltage (V_Z) is 5 to 8 Volts.	Avalanche breakdown is observed in the diodes with breakdown voltage larger than 8 Volts.
7.	Zener resistance is less. Since the doping level is high and the V-I curve is steeper.	Resistance is large since the doping is lower, and the V-I curve is not that sharp.

Table 2.2

A normal diode:

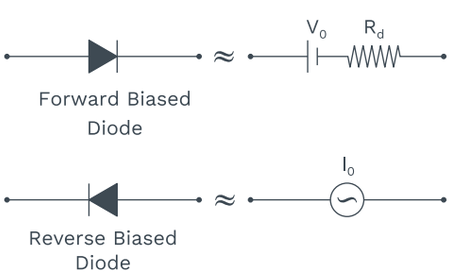


Fig. 2.76 Representation of Diode

- A Zener diode is a normal diode along with an extra region of operation, i.e. breakdown operation.

Here reverse voltage(V_R) > Zener voltage

Symbol:



Fig. 2.77 Representation of Zener Diode

Properties	Avalanche	Zener
Depletion region	Thick	Thin
Junction	Destroy	Not destroy
Electric field	Weak	Strong
Produces	Pair of electrons & holes	Electrons
Doping	Low	Heavy
Reverse potential	High	Low
Temperature coefficient	Positive	Negative
Ionization	Because of collision	Because of the electric field
Breakdown voltage	Directly proportional to temperature	Inversely proportional to temperature
After breakdown	Voltage varies	Voltage remains constant

Table 2.3

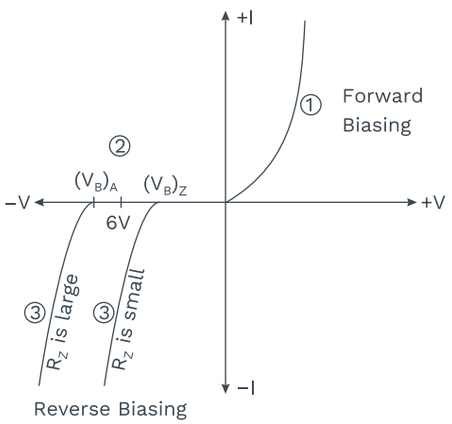


Fig. 2.78 V-I Curve for Zener and Avalanche Breakdown

where $(V_B)_Z$ = Zener breakdown voltage
$(V_B)_A$ = Avalanche breakdown voltage

- Overall, a Zener diode operates in three regions.

Region-1 and region-2 comprise operations just like a normal diode in forward and reverse bias, respectively.
The region-3 includes the special working region which is different and consists of working of a diode based on the magnitude of the breakdown voltage.

Note:
A normal diode will be burndown or damaged in the region-3.

- **Applications:**

Voltage regulator (Stabilizer):

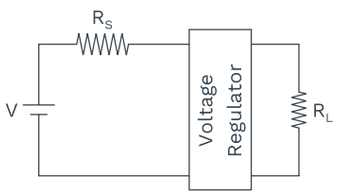


Fig. 2.79 Voltage Regulator

Rack Your Brain

The tunnel diode V-I graph shows negative resistance, so can a tunnel diode behave like a battery?

Rack Your Brain

What is the reason for increasing light intensity?

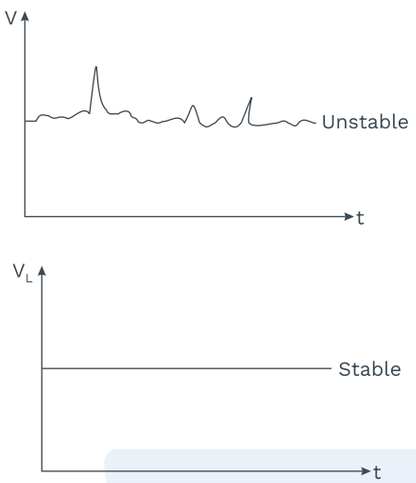


Fig. 2.80 Input and Output Voltage Characteristics of Voltage Regulator

- In place of a regulator, a Zener diode can be used.

Relationship between $(\vec{E}_{max})_B$ and V_B across the junction:

Applying Poisson's equation.

$$\frac{\partial V}{\partial x} = E_{max} = \frac{eN_D W_N}{\in} = \frac{eN_A W_P}{\in}$$

The voltage on the P-side will be:

$$V_{0P} = \frac{eN_A}{2 \in} W_P^2$$

and the N-side,

$$V_{0N} = \frac{eN_D W_N^2}{2e}$$

The breakdown voltage (V_B) is the total voltage:

$$\therefore \quad V_B = V_{0P} + V_{0N}$$

$$V_B = \frac{e}{2 \in}\left(N_D W_N^2 + N_D W_P^2\right)$$

$$V_B = \frac{e}{2 \in} N_D W_N \left(W_N + W_P\right)$$

$$\therefore \quad V_B = V_Z = V_0 - V_A$$

where, V_A = Applied Voltage

V_0 = Cut-in Voltage/ Buit-in Voltage

$$\therefore \quad V_B = V_Z = V_0 - V_A = \frac{E_Z}{2} \cdot W_B$$

Rack Your Brain

How photodiodes are different from solar cells?

Here,

$E_Z = E_{BD} = E_{max}$ {Electric field at breakdown}

W_B = Width of breakdown.

In the above equation, we still do not know the width at the breakdown.

- Thus, the relationship between the maximum electric field at the breakdown and breakdown voltage is –

$$V_B = \frac{E_Z}{2}.W_B$$

- **Relationship between; E_{max} and V_B; without W_B:**

We know:

$$V_B = \frac{E_{max}}{2} \cdot W_B$$

But also, the voltage across the diode (V_D) is:

$$V_D = \frac{2N_DN_A}{2 \in (N_D + N_A)} \cdot W^2 = V_0 - V_A$$

Substituting in the above equation:

$$\therefore \quad W_B = \sqrt{\frac{2 \in (N_D + N_A)}{e N_D N_A} \cdot V_B}$$

$$V_B = \frac{E_{max}}{2} \cdot \sqrt{\frac{2 \in (N_D + N_A)}{e N_D \cdot N_A} \cdot V_B}$$

$$V_B = \frac{E_{max}^2 \in}{2e}\left(\frac{1}{N_A} + \frac{1}{N_D}\right) = V_z$$

- The above equation explains that:

In Zener didoes, where N_D is large, the Zener voltage will be small.

In the avalanche diode, the N_D is small and the avalanche voltage will be larger.

Note:

Zener diodes are generally heavily doped, whereas the avalanche diodes are slightly doped.

Rack Your Brain

Why special construction is required in photodiode?

Rack Your Brain

The light sensitivity in forward biasing is _____.

	Diodes	Breakdown voltage
1.	Ordinary diode	150 – 400 Volts
2.	Avalanche diode	8 – 50 Volts
3.	Zener diode	3 – 8 Volts

Table 2.4

Zener diode as a voltage regulator:

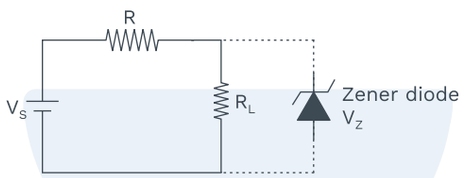


Fig. 2.81 Zener Diode as Voltage Regulator

The voltage across R_L is $V_{(R_L)}$, it is given as:

$$V_{(R_L)} = \left(\frac{R_L}{R + R_L}\right) \cdot V_s$$

By placing a Zener diode across R_L will help in stabilizing such as:

"Any voltage fluctuation in V_S will produce extra current across resistance (R) and which will flow across Zener diode, while maintaining a constant current across the R_L". (Provided $V_S > V_Z$)

Then, $V_{R_L} = V_Z$ {After using Zener diode}

- Without a regulator (Zener diode here), the voltage across is:

$$V_{R_L} = \left(\frac{R_L}{R + R_L}\right) V_s$$

However, V_S fluctuates or varies. ($\because$ Rectified output)

$\therefore$ V_{R_L} also varies & current through R_L, i.e. I_L will vary.

Given, $V_s > V_Z$ and Zener diode are working beyond the breakdown.

Any charges in V_s will change the current in the Zener diode keeping I_L and V_L constants.

Example 15:
Consider the following given circuit:

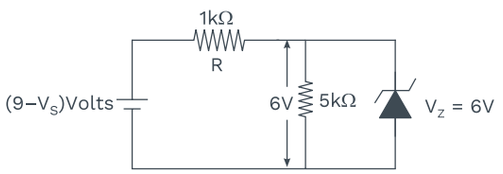


Fig. 2.82 Circuit Containing Zener Diode

If $I_{z(min)}$ = 0.1 mA & Zener resistance is zero.
Evaluate the $V_{s(min.)}$

Sol:
Given that,

$R = 1\ k\Omega$

$R_L = 5\ k\Omega$, $V_L = 6$ Volts

$V_Z = 6$ Volts, $I_{Z(min)} = 0.1$ mA

$$\therefore \quad I_L = \frac{6\,\text{Volts}}{5\,k\Omega} = 1.2\,mA$$

and $I_{R(min)} = 1.3$ mA
the voltage across resistance:

$\therefore \quad V_R = 1 \times 1.3 = 1.3$ Volts

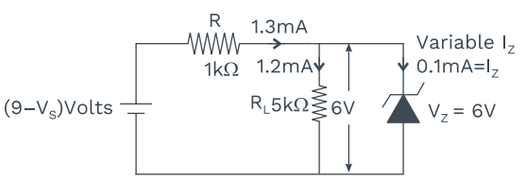


Fig. 2.83 Modified Circuit Containing Zener Diode

According to the circuit:

$9 - V_{S(min)} - 6 = V_R$

$9 - V_{S(min)} - 6 = 1.3$

$V_{S(min)} = 1.7$ Volts

Rack Your Brain

List some of the practical applications of the P-I-N diode.

Example 16:
Consider the circuit given where silicon diode and germanium diodes start conducting at 0.7 volts and 0.3 volts, respectively. The values V_0 and I, respectively are _____.

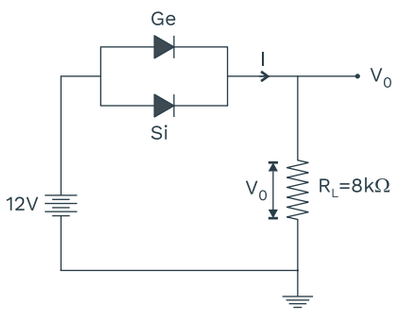


Fig. 2.84 Circuit Containing Diodes Fabricated from Ge and Si

a) 11.7 V, 1.46 mA **b)** 1.46 V, 11.7 mA **c)** 11.7 V, 1.23 mA **d)** 1.17 V, 146 mA

Sol:

From the given circuit, it is concluded that while forward biasing, the silicon diode will never be in the cut-in stage.

Because if 0.3V is achieved, then the germanium diode gets into ON state.

And once the germanium is in conducting state, its voltage across it will be 0.3 volts & the silicon diode will not get 0.7 volts as per its requirement.

$\therefore$ V_{Ge} = 0.3 volts (ON state)

V_{Si} = 0.3 volts (OFF state)

$\therefore$ The Voltage across R_v will be

$12 = 0.3 + V_D$

V_0 = 11.7 volts

& $I = \dfrac{11.7\ \text{volts}}{8\,\text{k}\Omega}$

I = 1.46 mA

Hence, the correct option is a)

TUNNEL DIODE

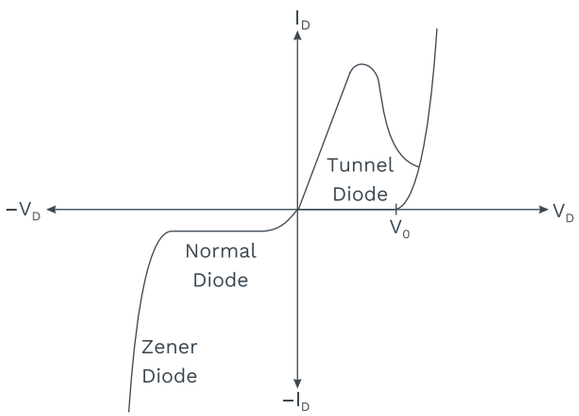


Fig. 2.85 Tunnel Diode

- A tunnel diode can conduct in forward bias before the cut-in voltage is applied. The region is referred to as the behaviour region.

Symbol:



Fig. 2.68 Tunnel Diode

There are two different zones of operation in the tunnelling region:

i) An increasing voltage along with the increasing current.

ii) An increasing voltage and decreasing current.

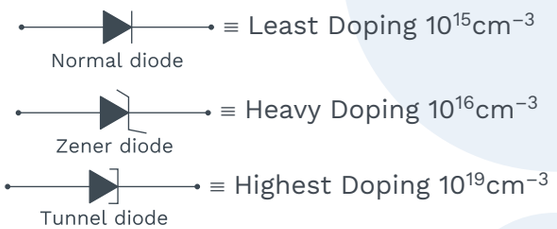


Fig. 2.87 Doping Concentration in Various Diode

- Heavy doping normally results in a reduction of the depletion regions width where the wave nature of the electron becomes predominant.
- Electrons travel the depletion region in an easier way than the normal particle behaviour; since the width of the depletion region is very narrow, the electron's wave nature can be seen easily.
- A tunnel diode is used in forward biasing.
- Tunnel diode conducts in forward bias before cut-in voltage.
- The special feature is that it conducts with increasing current first, then decreasing current before the cut-in voltage.
- The doping level in the tunnel diode is exceptionally very high.

  $E_F > E_C$ in N-side

  $E_F < E_V$ in P-side

Rack Your Brain

What is photo-luminescence?

Rack Your Brain

The recombinations occurring in LEDs are radiative or non-radiative?

The width of the depletion region is negligibly strong.

- The high doping ensures the narrow width of the depletion region.

Since,

$$V - V_D = \frac{e}{2 \in (N_D + N_A)} \cdot W^2$$

If $N_D >> N_A$

Then, $V_0 - V_D = \frac{eN_DN_A}{2 \in N_D} \cdot W^2$

$$W \propto \frac{1}{\sqrt{N_A}}$$

Schrodinger wave equation:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

- If V > E, the wave can become a particle that has exponential behaviour.
- At sub-atomic levels, if V > E, we get exponential behaviour, whereas if V < E, we get the wave function.

Thus, the overall behaviour of an electron depends on kinetic energy (E) & its potential (V).

- Many times, kinetic energy will decide the behaviour, while other behaviour will be decided on the basis of V.

This duality of behaviour is referred to as the dual nature.

- At 1Å unit of depletion region (narrow width), the electrons wavelength is comparable, and hence penetration is possible for the electron as a wave.

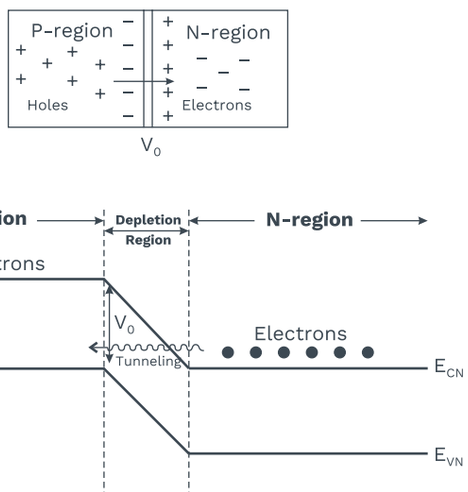


Fig. 2.88 Tunnel Diode

- If the applied bias voltage is less than the V_0 and positive forward bias, then the electron in N-region crosses over to P-region. i.e., E_{CN} to E_{VP} (for small $V_B < V_0$).

On increasing V_B (Bias voltage), the voltage (V_D) and diode current (I_D) both will increase, and finite resistance will be there.

- When $V_D < \frac{V_0}{2}$ and $V_D < V_0$.

Then, this makes the (E-V) terms to be negative.

If the (E-V) term becomes negative, then the electron will behave as a particle.

The (E-V) term is positive for small diode voltage (V_D), then E > V, the electron will behave as a wave.

Therefore, particle nature initiates and stops tunnelling at ($E < V_0$).

Thus, I_D will drop & V_D will increase & V_B will also increase.

$$I_D\downarrow\ V_D\uparrow\ V_B\uparrow$$

- To summarize:

If $V < \frac{V_0}{2}$, tunnelling is easy (Wave nature dominates)

If $V > \frac{V_0}{2}$, tunnelling stops (Particle nature dominates)

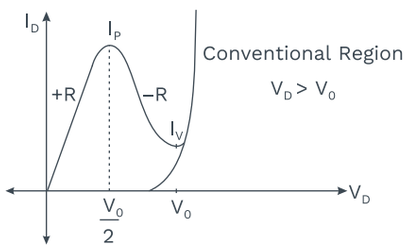


Fig. 2.89 V-I Characteristic Graph of the Tunnel Diode

Rack Your Brain

The LEDs are chromatic or non-chromatic in nature?

From the graph, it is seen that the tunnel diode behaves like a normal diode after cut-in voltage (V_0), but before that, it shows a tunnelling effect.

I_P is the peak current and I_V is the valley current.

The positive resistance region is where the current increases with the voltage (denoted as (+R) in the graph).

The negative resistance region is where the current decreases with an increase in voltage (denoted as (–R) in the graph).

- The key feature of the tunnel diode is that on increasing the voltage up to a certain level, the current will decrease.
- Every tunnel diode has a negative resistance region (–R) where if V_D increases, then I_D decreases.

Rack Your Brain

Why LED work only in forward bias?

Applications:

1) Tunnel diode can be used as negative resistance (–R) oscillator (DC to AC converter).

2) With positive $\left(\frac{+V_0}{2}\right)$, the diode current ($I_D$) is high.

- For positive ($+V_0$), the I_D is high, and for $V_0 = 0$, the I_D is low.

Thus, a tunnel diode is used for high-speed switching applications.

The tunnel diode act as a negative resistance $\left(\frac{+V_0}{2} \text{ to } V_0\right)$ with the diode capacitance (C_D) for $V_D < V_0$ is shown below:

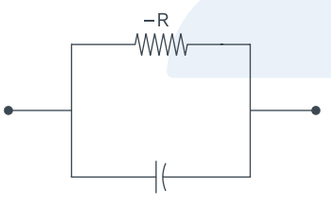


Fig. 2.90 RC Oscillator Circuit

Note:

If any device has positive resistance (+R), i.e. V-I are in phase, they always dissipate power.

For negative resistance (–R) where V-I are out of phase, there will be only energy shifts from one form to another and no power dissipation.

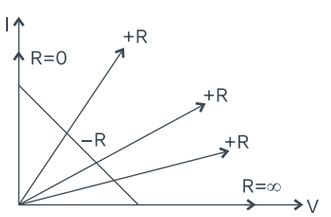


Fig. 2.91 Negative Resistance

SCHOTTKY DIODE

Schottky diode consists of metal-semiconductor junction instead of ordinary P-junction and N-junction region.

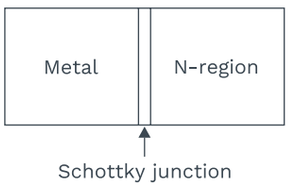


Fig. 2.92 Schottky Diode

- Since one side is metal, there is no recombination time, unlike the P-N junction where recombination occurs.

Schottky diode is specially designed to reduce the junction capacitance and switching times, and it is done by reducing recombination time and by minimizing the barrier potential, the recombination time can be reduced.

- Schottky diode is made up of metal-semiconductor (typically N-type) so that the metal conducts with electrons and the semiconductor also conducts with electrons.
- The advantage of this diode is that there is no need for recombination time requirements.

This reduced recombination time & reduced junction capacitance feature is used in high-speed switching.

It is also called a unipolar device, i.e. it only uses electrons.

Thus, the electron moves entirely in the conduction band, not in the valence band.

Energy-band diagram:

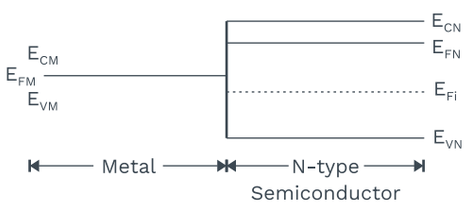


Fig. 2.93 Energy-Band Diagram of Schottky Diode

In the metal region, E_{CM}, $E_{VM,}$ and E_{FM} are almost overlapping, as can be observed in the figure. As soon as the junction forms, the diffusion of electrons will start, flowing from N-region to the metal, as seen with the $E_{FN} > E_{FM}$. Due to this, the junction will create positive ions; thereby, energy (E_{CN}) will rise above.

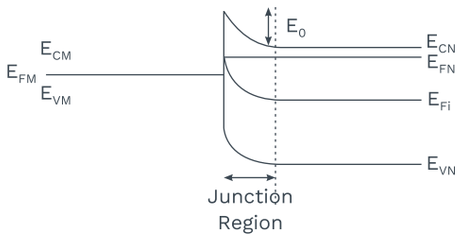


Fig. 2.94 Energy-Band Diagram of Schottky Diode at the Junction

- The Fermi level (E_{FN}) on N-side is close to the E_{CN} and also with E_{VN}.

But near the junction region, the E_{FN} is away from E_{CN} and E_{VN}.
This is called as E_0 barrier, or the Schottky barrier develops across the junction.
There is no carrier accumulation; thus, there is no junction capacitance.

- From N-region to the metal region conduction, electrons will experience the Schottky barrier.

And from metal to N-region, there will be no conduction or no movement in open-circuit conditions ($E_{FM} < E_{CN}$).
In other words, from N to the metal region, electron movement will require V_B (Barrier voltage). Metal to N-region will need $(E_{FM}–E_{FN}) + V_B$ voltage & overall no chance of conduction. The depletion region will be only on N-side ($E_{FM} < E_{FN}$).

- **Ohmic contact:** It is referred to as the metal and N-type semiconductor contact where ($E_{FM} = E_{FN}$) and no barrier potential or no junction drop.

If $E_{FM} > E_{FN}$ then:

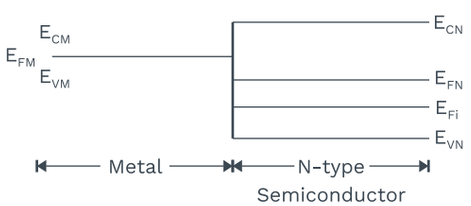


Fig. 2.95 Energy-Band Diagram of Schottky Diode

In this case, there will be diffusion of electrons in the metal to semiconductor resulting in negative ions in the N-region forming a depletion region.

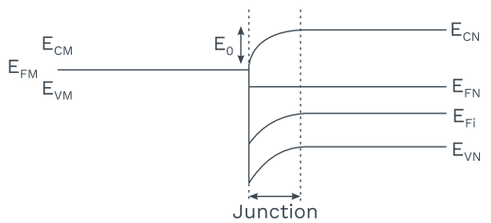


Fig. 2.96 Energy-Band Diagram of Schottky Diode as Ohmic Contact

E_0 = Barrier potential.

- Reduced recombination time causes a reduction of junction capacitance along with reduced forward voltage drop.

Clippers and rectifiers need the reduced forward voltage drop.

- The extreme case of the Schottky diode is the point-contact diode.

A Schottky barrier diode of reduced junction content area is referred to as the point-contact diode.

V-I characteristics:

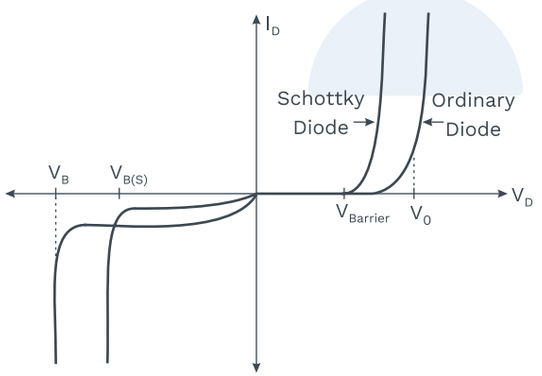


Fig. 2.97 V-I Characteristics of Schottky Diode

where

$V_{Barrier}$ = Barrier voltage of Schottky diode

$V_{B(S)}$ = Breakdown voltage of Schottky diode

V_B = Breakdown voltage

V_0 = Cut-in voltage of a normal diode

- The graph predicts that:

A Schottky diode has lower forward barrier potential voltage and lower reverse barrier voltage ($V_{B(S)}$)

SOLAR CELL – PHOTOVOLTAIC CELL

- It is also named Photovoltaic cell (PV cell). On considering the energy band diagram of a normal P-N junction diode, if P-region or the N-region can have minority carriers initiated in the diode in forward voltage condition before cut-in voltage (open circuit), then there will be a huge current by flowing electrons from P-region to N-region.
- In an open circuit and with no external voltage applied, it is capable of breaking bonds and producing carriers, the large value of current will flow in the reverse bias. (called reverse current).

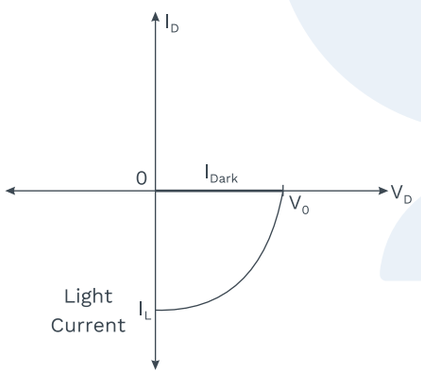


Fig. 2.98 V-I Characteristics of Photovoltaic Cell

Case-I: Open-circuit conditions, i.e. no external voltage (zero bias voltage and light is focused on the diode).

V_D = 0V, there will be current due light, i.e. I_L

It is in IVth quadrant since the current is flowing due to P-region electrons to N-region (reverse current).

When open circuit conditions are assumed with the incidence of high on P-region. Minority carriers are produced, i.e. a reverse current initiates.

In other words, that current is flowing due to light incidence.

When the solar cell voltage is not V_0, it has a dark current (I_D). But when the voltage reaches V_0. The electrons will start to move from N-region (E_{CN}) to P-region (E_{CP}), and then the current will drop.

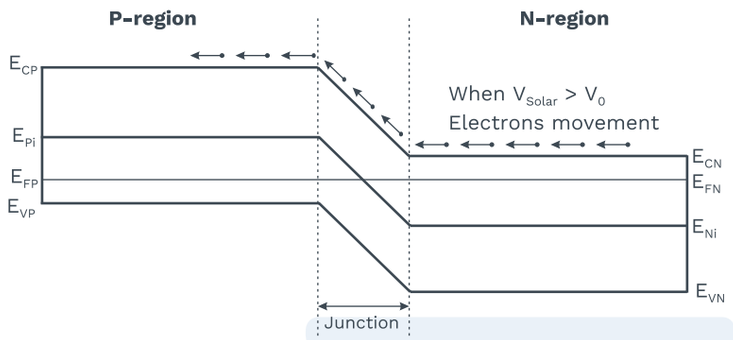


Fig. 2.99 Movement of Electrons in Energy Bands of Photovoltaic Cell

If this continuously flows, increasing diode junction voltage to V_0 light current (I_L) will drop or decrease to zero.

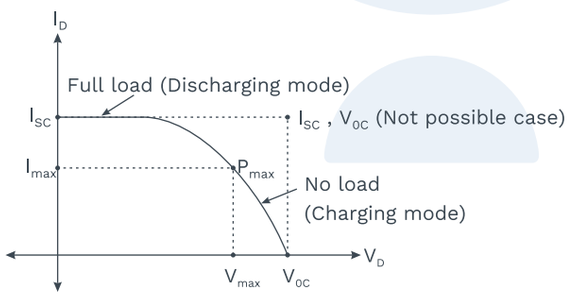


Fig. 2.100 V-I characteristics of Photovoltaic Cell

- Solar cell is identified by 2-stages:
  - **i)** Heavy current stage
  - **ii)** Zero current stage

Solar cell is used in such a manner that current and voltage have large (maximum) magnitude so that it has maximum power across them.

It is not possible to have I_{SC} and V_{0C} at the same time, but only it is observed that I_{max} (finite voltage) is possible to achieve in order to get maximum power output from the solar cell.

Symbol:



Fig. 2.101 PV Cell System

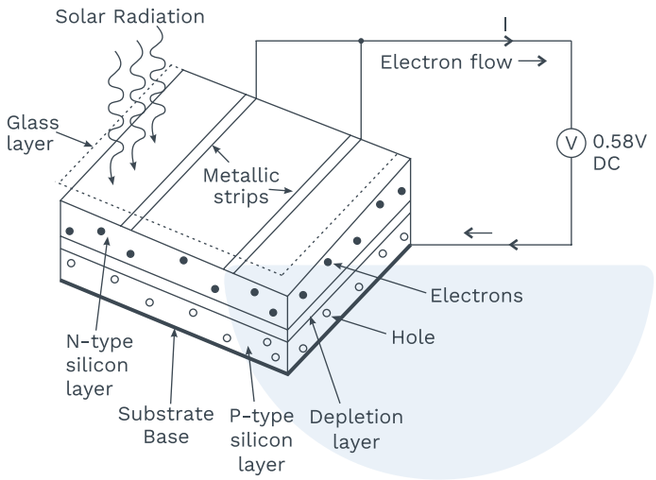


Fig. 2.102 Working of Photovoltaic Cell

- Solar cells can produce 0.7V, but practically, it gives 0.58V.

Thus, solar modules are mainly in a solar module system (panel).

Any active device that generates power has a property of voltage decrease on increasing current (V↑ I↓).

If I_{SC}, then V = 0V and

If I_0, then V = V_{0C}

This same phenomenon occurs in solar cells that, on drawing heavy current then, its voltage will drop across it.

- The V-I characteristic defines that I = 0A for V = V_{0C} = V_0 (Cut-in voltage) and current is maximum I_{max} or I_{SC} for V = 0V.

Such that:

$$V_{0C} \times I_{SC} = P \text{ (Hypothetical maximum power)}$$

Practical maximum power

$$P_{max} = V_{max} \times I_{max}$$

$$\text{Fill factor} = \frac{V_{max} \times I_{max}}{V_{0C} \times I_{SC}}$$

- **Fill factor:** Ratio of the area occupied with the maximum power to the total area. For any solar cell, the fill factor is an important criterion for designing aspects.

LIGHT DEPENDENT RESISTANCE (LDR)

- They are also called a photodiode. The special case of this is also called an avalanche photodiode (APD) and P-I-N (PIN diode).
- The central principle for the working of the photodiodes is that they only start to conduct when applied with the bias and light is incident on them.

They are all considered passive switches that are used in reverse bias to conduct light and act as closed switches.

In the darkness, they behave as an open switch.

- A normal photodiode (or LDR) works on a principle of reverse current increasing with respect to the intensity of light incidence.

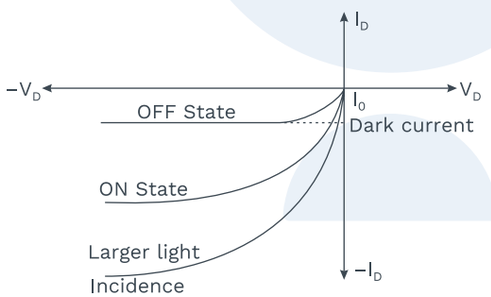


Fig. 2.103 V-I Characteristics of LDR

Working principle:

The larger the light incident on the diode, causes the larger amount of broken bonds, and therefore the minority currents are produced.

With E = hf

E = Energy of light

f = Frequency of light

This energy causes a high amount of bond breakage in reverse bias, and hence a large sum of minority carriers is produced.

This causes an off device to start conducting.

- **APD (Avalanche photodiode):**

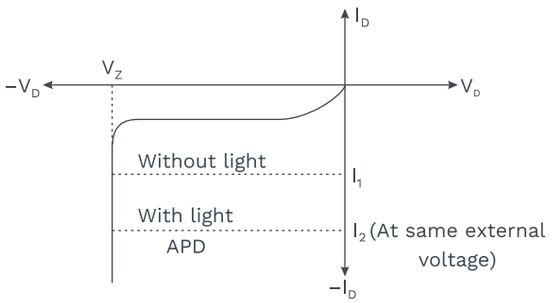


Fig. 2.104 V-I Characteristics of APD

While operating at the same external voltage, if the diode produces a large current with the incidence of light, it is called APD (Avalanche Photodiode.)

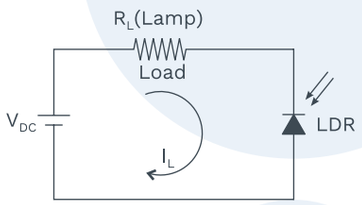


Fig. 2.105 LDR Connected to Load

The above circuit is an example of an application.

- In this case, LDR will produce I_L (light current) on the light incidence and I_{Dark} when there is no incident. This shows the switching of LDR.

But if an APD is used instead of LDR, it must have V_{DC} equal to the breakdown voltage, i.e.

$$V_{DC} = V_B$$

Then only the switching circuit will be the same. The current flowing is breakdown current ($I_L = I_{Breakdwon}$) for APD while conduction during an incidence of light.

APD- switching circuit:

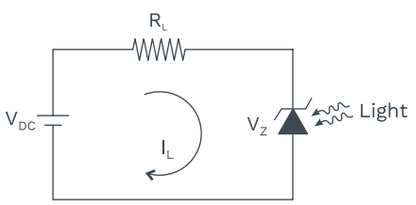


Fig. 2.106 Switching Circuit Using APD

$V_{DC} > V_Z$ such that I_L should flow if the light intensity is high.

If $V_{DC} \cong V_Z$ (close to breakdown condition)

Then without light I_L = 0A, and with light, I_L will increase drastically.

Application of APDs and LDRs:

- Used especially in street light and light-based controlled devices which has high sensitivity.
- For lesser sensitivity, normal LDR photodiodes are used.
- Photodiode APDs in forward biasing do not have not have any additional advantage.

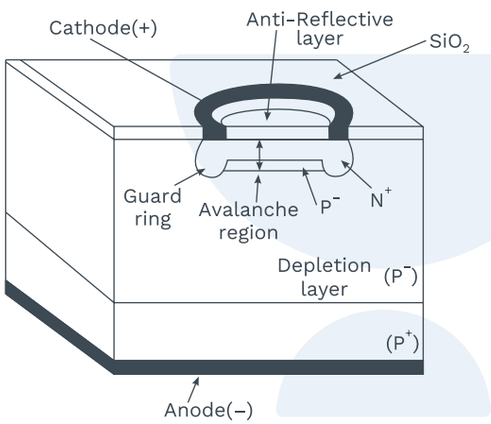


Fig. 2.107 APD

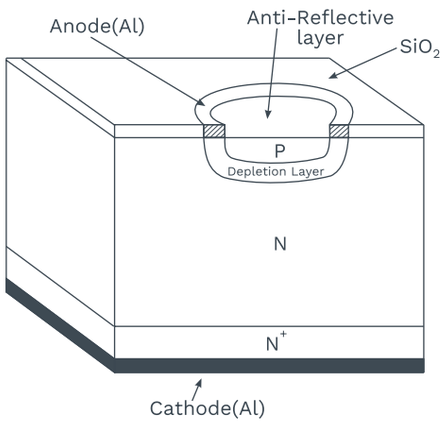


Fig. 2.108 LDR

- In APD construction, there is a slightly doped region present additionally between N & P-region.
- In APD, the P-N junction has a prominent region that is lightly doped.

This additional region is required because minority carriers are produced near the junction only, and this layer is needed to maintain, and produce minority carriers, and this is why it is lightly doped.
The depletion region is very large in the APDs.

- And if we take the intrinsic junction region in between the N-region and P-region, the diode becomes a P-I-N diode.

This intrinsic junction is very slightly doped (almost pure).
This reduced doping in the junction supports minority carriers and reduces recombinations.
The produced minority carriers due to light should be close to the junction for successful cross-over to the other side.

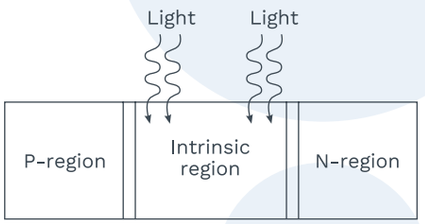


Fig. 2.109 PIN Diode

Therefore, P-I-N supports a large amount of cross-over of minority carriers with the least amount of recombinations.

- Instead of light, sometimes the microwave is incident on the intrinsic region, and it also starts conducting. In a radar system, microwave sensing is done using P-I-N diodes.
- A long intrinsic or lightly doped region between P^+ and N^+ regions ensures a sufficiently strong reverse current.
- P^+ P P^- N^- N N^+ type of construction is also used and called read diode.

LED (LIGHT-EMITTING DIODE)

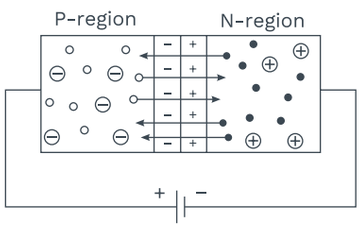


Fig. 2.110 LED

- Considering the above circuit, in forward biasing where holes form, the P-region will move to the N-region and electrons from N-region will move to the P-region.

Holes get recombined with electrons in the N-region, and thereby the holes concentrations decrease.

Similarly, the electrons get recombined in P-region and thereby, the electron's concentrations will decrease in the P-region.

Since it can be observed that recombinations are occurring in both regions. And it is known that after the recombination process, there will be definitely the generation of heat or light depending upon the direct or indirect bandgap of the semiconductor.

- The P and N-type regions are made up of GaAs.

Symbol:



Fig. 2.111 LED

- In LEDs, the recombination process is radiative in nature.
- Since there is also an injection of charged particles like holes are injecting from P to N-region and electrons injecting from N to P region because of an external electric field.

This recombination process produces light due to these injections of carriers.

Thus, this whole procedure is referred to as the injection luminance.

This external electric field is responsible for the injection of charge carries and subsequent recombination in generating light. The whole process is known as electroluminescence.

- In LED, spontaneous emission occurs.

Advantages:

i) Light weight
ii) Small size
iii) Long life
iv) Rugged constructions
v) Low power consumptions.

Disadvantages:

i) LEDs are not highly directional.
ii) Not chromatic (not a single wavelength)

Injection efficiency of LED:

- The injection efficiency of a P-N junction is defined for one-side P-N junctions (where one of the regions is highly doped than the other regions).
- Injection efficiency in a forward bias P-N Junction is the amount of current produced due to one-type charge particle out of the total current.

If the P^+N junction is considered, (where this (+) sign is denoted by the additional doping amount).

$\therefore$ Here P-region is doped more than the N-region.

$\therefore$ Injection efficiency will be:

$$\eta = \frac{I_{PN}(0)}{I_{NP}(0) + I_{PN}(0)}$$

$I_{PN}(0)$ = Injected current from N to P side at x = 0

Also,

$$\eta = \frac{Ae\frac{D_P}{L_P}P_{N0}\left(e^{V/V_T} - 1\right)}{Ae\frac{D_P}{L_P}P_{N0}\left(e^{V/V_T} - 1\right)} + Ae\frac{D_N}{L_N}N_{N0}\left(e^{V/V_T} - 1\right)$$

$$\eta = \frac{\frac{D_P}{L_P}P_{N0}}{\frac{D_P}{L_P}P_{N0} + \frac{D_N}{L_N}N_{P0}}$$

and we know that:

$$P_{N0} = \frac{n_i^2}{N_D}, \qquad N_{P0} = \frac{n_i^2}{N_A}$$

$$\eta = \frac{\frac{D_P}{L_P}\cdot\left(\frac{1}{N_D}\right)}{\frac{D_P}{L_P}\left(\frac{1}{N_D}\right) + \frac{D_N}{L_N}\left(\frac{1}{N_A}\right)}$$

And $L = \sqrt{D\tau}$

$$L_P = \sqrt{D_P\tau_P}$$

$$L_N = \sqrt{D_N\tau_N}$$

$$\therefore \quad \eta = \frac{\sqrt{\frac{D_P}{\tau_P}}}{\sqrt{\frac{D_P}{\tau_P}}\left(\frac{1}{N_D}\right) + \sqrt{\frac{D_N}{L_N}}\left(\frac{1}{N_A}\right)}$$

V-I characteristics:

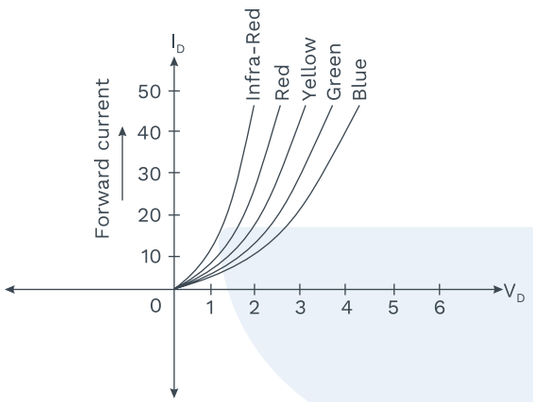


Fig. 2.112 V-I Characteristics of LED

Chapter Summary



- At the centre of the depletion layer, the electric field:

  $$E_{max} = \frac{eN_D W_N}{\in} = \frac{eN_A W_P}{\in}$$

  At any distance, $E(x) = \frac{eN_D(W_N - x)}{\in} = \frac{eN_A(W_P - x)}{\in}$

- At P-region, $V_{0P} = \frac{eN_A W_P^2}{2 \in}$

  At N-region, $V_{0N} = \frac{eN_A W_N^2}{2 \in}$

  $$V_{Total} = V_0 = \frac{eN_A N_D W_N^2}{2 \in (N_A + N_D)} = V_{0P} + V_{0N}$$

  $$V_0 = \frac{e}{2 \in}\left(N_A W_P^2 + N_D W_N^2\right)$$

- In forward bias or reverse bias:

  $$(V_0 - V_{(x)}) = \frac{eN_D N_A}{2 \in (N_D + N_D)} x^2$$

  {$V < V_0$ for forward bias and V can be any voltage at reverse bias}

- Cut-in voltage, $V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$

- Law of junction, $V_0 = V_T \ln\left(\frac{P_P}{P_N}\right) - V_T \ln\left(\frac{N_N}{N_P}\right)$

- $I = I_0\left(e^{V/\eta V_T} - 1\right)$

  $\eta = \begin{cases} 2 \text{ for Silicon} \\ 1 \text{ for Germanium} \end{cases}$

- Diode capacitance (C_D):

  $$C_D = \frac{\tau}{r_D} = \frac{I \cdot \tau}{\eta V_T} = \frac{Q}{V}$$

  $$r_D = \frac{\eta V_T}{I} = \frac{dV}{dI}$$

- Transition capacitance (C_T):

  $$C_T = \frac{\in A}{W}$$

PRACTICE QUESTIONS

Q1 **Consider a solar cell with a 2 cm^2 area and operating at 0.1 sun intensity. The short circuit current is 2.6 mA, and the open-circuit voltage is 0.81 volts. Determine the open-circuit voltage at sun intensity of 0.3. Assuming the room temperature operation and thermal equivalent voltage of 26 mV.**

a) 0.778 volts
b) 0.8 volts
c) 0.678 volts
d) 7.7 volts

Q2 **Find V_0 in the given circuit.**

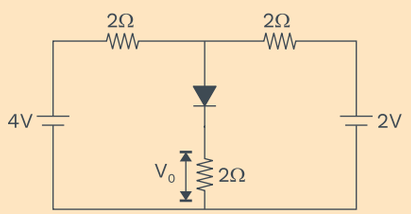


Fig. 2.113 Circuit Containing Diode

a) 0 volts
b) 2 volts
c) -2 volts
b) 1 volts

Q3 **Find V_0 in the given circuit.**

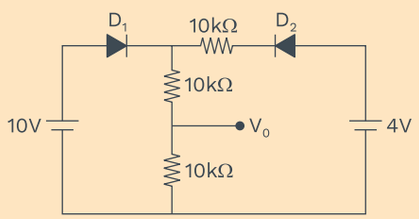


Fig. 2.114 Circuit Containing Diode

a) 5 volts
b) 0 volts
c) 2 volts
d) 2.5 volts

Q4 **Find V_0 in the given circuit:**

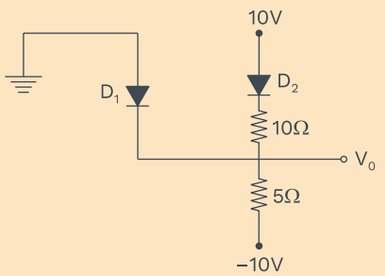


Fig. 2.115 Circuit Containing Diode

a) 1.2 Volts **b) 0.1 Volts**
c) 3.3 Volts **d) 0 Volt**

Q5 **The resistance across A and B is:**

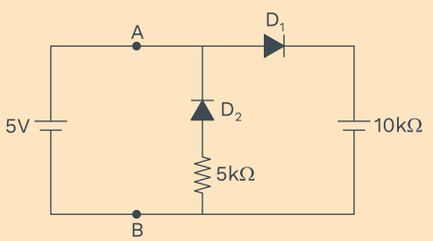


Fig. 2.116 Circuit Containing Diode

a) 1 kΩ **b) 5 kΩ** **c) 10 kΩ** **d) 2.5 kΩ**

Q6 **The doping concentrations in the P-region and N-region are $N_A = 7.91 \times 10^{16}$ cm^{-3} and $N_D = 1 \times 10^{16}$ cm^{-3} for an abrupt P-N junction.**
If the P-N junction is reversed biased and the total depletion width is 2.75 μm. The depletion width on P-side is:

a) 0.25 μm **b) 1 μm** **c) 0.308 μm** **d) 3.08 μm**

Q7 Consider a silicon P-N junction diode under reverse biasing with a 7.45 μm depletion region width. Evaluate the diode capacitance of the diode/cm^2 if the relative permittivity of silicon is 11.7?

a) 13.89 μF b) 12 F c) 13.8 F d) 20 μF

Q8 If a silicon PN junction at a temperature at 20°C has 20 pA reverse saturation current, when will the reverse saturation current be 120 pA?

a) 30°C b) 40°C c) 60°C d) 120°C

Q9 The built-in potential of the P-N junction is 1.2V. If the depletion layer width at reverse bias voltage 1.8V is 4 μm. What will be the depletion layer width at a reverse bias of 7.5 volts?

a) 6.8 μm b) 2.6 μm

c) 5.8 μm d) 6.8 m

Q10 A silicon P-N junction with a doping concentration of acceptor as 1.2×10^{17} cm^{-3} on P-region and donor concentration as 1.4×10^{16} cm^{-3} in N-region.

Assume that no external voltage is applied to the diode.

The charge in the depletion region on the P-side (in nC cm^{-2}) is _____

Assume, kT/q = 26 mV

$n_i = 1.5 \times 10^{10}$ cm^{-3}

$\in_0 = 8.85 \times 10^{-14}$ F/m

$q = 1.6 \times 10^{-19}$ C

a) –1.8 b) 1.8

c) –2 d) 1

Q11 **Consider a silicon device fabricated in $n^+ - n$ configuration with a uniform doping donor concentration of 1.3×10^{18} cm^{-3} and a non-degenerate donor concentration of 1.2×10^{14} cm^{-3}, respectively.**
Assume complete impurity ionization at room temperature, kT/Q = 26 mV and $n_i^2 = 10^{10}$ cm^{-3}. The built-in potential of the given device is:

a) 0.671 V **b) 0.173 V** **c) 0.71 V** **d) 0.134 V**

Q12 **Consider a P-N junction that is of holes and electrons. And assume that ionized donor density $N_D^+ = 10^{17}$ cm^{-3}. The electric field at n = 0 is 0.**
At what length (l), the electric field will be σ 75 kV/cm?
Assume that $\in_0 = 8.854 \times 10^{-14}$ F/cm and $\in_r = 11.7$ for silicon.

a) 2.4 μcm **b) 4.8 μcm** **c) 4.8 nm** **d) 1.2 μcm**

Answer key:

1) a)	**2) c)**	**3) a)**	**4) d)**	**5) c)**	**6) c)**
7) a)	**8) a)**	**9) a)**	**10) a)**	**11) a)**	**12) b)**